

Approx. 1971

MACQUARIE UNIVERSITY

School of Mathematics and Physics

09316 - TOPICS IN GEOMETRY

Supplementary Lecture Notes prepared by R. Street on

SYNTHETIC PLANE PROJECTIVE GEOMETRY

Copyright (other than in works quoted) reserved by Macquarie University.

MACQUARIE UNIVERSITY

School of Mathematics and Physics

09316 - Topics in Geometry - R.H. Street

Text: B.E. Meserve, *Fundamental concepts of geometry*, Addison-Wesley.

The purpose of this course is to study various geometries of the (real and complex) plane. Since the various geometries in which we are interested are subgeometries of the projective plane, the approach is to study projective plane geometry in some detail.

Notions such as *similarity* and *congruence* will be considered and made precise. These notions were considered by Euclid, but the lack of precision in Euclid's definitions and axioms should be apparent (see Meserve Chapter 7, especially 7-3). Two figures which are congruent are certainly similar, so, in a sense, similarity is a weaker form of equivalence between figures than congruence. The study of those properties of figures which are common to all similar figures is called *Euclidean similarity geometry*. The study of those properties of figures which are common to all congruent figures is called *Euclidean metric geometry*.

There is an equivalence between figures which is even weaker than similarity. This may appear too weak to be of any importance since any two triangles are equivalent, any ellipse is equivalent to the unit circle, and even ellipses and parabolae and hyperbolae are equivalent. However, there are properties which all triangles do have in common and there are properties which all conics (ellipses, hyperbolae and parabolae) do have in common. So the study, *projective geometry*, of those properties of figures which are common to all figures equivalent in this way is still worth considering.

The following quote concerning projective geometry comes from S.H. Gould. "The fundamental idea for this pure geometry came from the desire of Renaissance painters to produce a 'visual' geometry. How do things really look, and how can they be presented on the plane of the drawing? For example, there will be no parallel lines, since such lines appear to the eye to converge". If a painting of a circle is held slightly on its side it appears as an ellipse. A painting of an equilateral triangle can appear to be any shaped triangle when viewed from different angles. The equivalence of figures considered in projective geometry can intuitively be thought of as follows:

two finite figures in the plane are equivalent if one of the figures can be viewed from some angle so that it looks like the other one.

This, as a definition, is even less precise than Euclid's definitions of similarity and congruence. Notice, however, if three points are on the same line (the three points are *collinear*) then this remains so no matter how we view the page; and if three lines pass through the same point (the

three lines are *concurrent*) then this also remains so no matter how we view the page. We will show in this course how the fact that these properties of concurrency and collinearity are preserved no matter how the plane is viewed leads to a precise definition of when any two figures are *projectively equivalent*, and eventually, of when they are similar or congruent.

The basic concepts necessary for a projective plane then are points, lines and a formula which tells when a given point and a given line are incident. There should be exactly one line which is incident with two given distinct points as in Euclidean plane geometry. In Euclidean plane geometry, two distinct lines are incident with exactly one common point *unless the lines are parallel*. Parallelness leads to exceptional cases in the statements of many theorems. An extra *ideal line* (or line at infinity) can be added to the Euclidean plane so that parallel lines meet on this ideal line. Then the Euclidean plane plus the ideal line has the property that *any* two distinct lines meet in exactly one point.

CHAPTER I: AXIOMATIC PROJECTIVE GEOMETRY

§1. Projective planes

A *projective plane* consists of a set whose elements are called *points*, a set whose elements are called *lines* and a relation, called *incidence*, between the points and the lines, and the following axioms must be satisfied:

1. *there is exactly one line which is incident with two distinct points;*
2. *there is at least one point which is incident with two distinct lines;*
3. *there exist four distinct points, no three of which are incident with any one line.*

In order to have a projective plane then the points, the lines and the relation of incidence must be given, and the three axioms must be verified.

If P is a point and ℓ is a line, instead of saying P and ℓ are incident we may also say P is on ℓ , P lies on ℓ , ℓ goes through P , etc. If P, Q are two distinct points then axiom 1 says there is exactly one line which goes through both P and Q ; this line is denoted* by $P \vee Q$ (P join Q). If R lies on $P \vee Q$ then we say that P, Q, R are *collinear*. So axiom 3 says that there are four points no three of which are collinear.

Proposition: *There is exactly one point which is incident with two distinct lines.*

Proof: Suppose ℓ, m are distinct lines. There is at least one point P on both ℓ and m by axiom 2. Suppose Q is on both ℓ and m . If P, Q are distinct then we have two lines ℓ, m through two distinct points contrary to axiom 1. So $P = Q$.//

The unique point which lies on both the lines ℓ and m when ℓ, m are distinct is denoted* by $\ell \wedge m$ (ℓ meet m). If n goes through $\ell \wedge m$ then we say that ℓ, m, n are *concurrent*.

Proposition: *There exist four distinct lines no three of which are concurrent.*

Proof: By axiom 3 there are four distinct points $P_i, i = 1, 2, 3, 4$ no three of which are collinear. Then the four lines $P_1 \vee P_2, P_2 \vee P_3, P_3 \vee P_4, P_4 \vee P_1$, are distinct (since no three of the P_i are collinear) and no three are concurrent (since P_1, P_2, P_3, P_4 are distinct).//

* Note difference with Meserve.

Proposition: *Every line is incident with at least three points.*

Proof: Let ℓ be any line. There exist four distinct points P_1, P_2, P_3, P_4 no three collinear. No more than two of these points can be on ℓ . Suppose P_1 is not on ℓ . Then the three points $(P_1 \vee P_2) \wedge \ell$, $(P_1 \vee P_3) \wedge \ell$, $(P_1 \vee P_4) \wedge \ell$ on ℓ are distinct. //

The set of points on a line is called a *pencil of points* and the set of lines through a point is called a *pencil of lines*.

§2. Duality

Every *statement* about projective planes can ultimately be phrased in terms of points, lines and incidence. If we replace the word "point" by "line" and the word "line" by "point" at every place they appear in the statement then a new statement, called the *dual statement*, is obtained.

A *concept* in projective geometry is defined by a statement. The concept defined by the dual statement is called the *dual concept*. For example, the statement in italics defines "lies on":

a point lies on a line when the point and line are incident.

The dual of this statement in italics is the following statement in italics:

a line goes through a point when the line and point are incident,

and this latter is the definition of "goes through". So the dual concept to "lies on" is "goes through".

If a statement involves various concepts, then to obtain the dual statement there is not always need to rephrase the statement purely in terms of points, lines and incidence when the dual concepts are known; rather, all the concepts should be replaced by their dual concepts.

The following statement in italics defines "collinear":

a set of points are collinear if they lie on a common line.

The following statement in italics defines "concurrent":

a set of lines are concurrent if they go through a common point.

The statements in italics are dual, so the dual concept of "collinear" is "concurrent".

The three axioms themselves are statements.

The dual statements are:

dual 1. there is exactly one point which is incident with two distinct lines;

dual 2. there is at least one line which is incident with two distinct points;

dual 3. there exist four distinct lines no three of which are incident with any one point.

In the last section we saw that these three statements are true (that is, are consequences of the axioms).

This leads us to the metamathematical rule:

Principle of duality: *If a statement about all projective planes is true, then so is the dual statement.*

The logical principle involved is that, since the duals to all the axioms are consequences of the axioms, the dual to any consequence of the axioms is also a consequence of the axioms.

There is a purely mathematical reason why the principle of duality holds. Given a projective plane P we can define P^* . The points of P^* are the lines of P and the lines of P^* are the points of P , and incidence occurs between a point and a line of P^* exactly when the corresponding line and point of P are incident. The axioms 1, 2, 3 for P^* are exactly the duals 1, 2, 3 for P , and these are true since P is a projective plane. So P^* is a projective plane, called *the dual projective plane* of P . A statement about P^* is true if and only if the dual statement about P is true. So if a statement is true about all projective planes P then it is true about all dual projective planes P^* (since P^* is a projective plane), and hence the dual statement is true about all projective planes P . This "proves" the principle of duality.

Note that the dual statement of the dual of a statement is the original statement. Two statements are said to be *dual* if one is the dual of the other. Symbolically, $P^{**} = P$.

§3. Examples

One of the purposes of this course is to put Euclidean plane geometry on a firm foundation. No doubt the student has an intuitive idea of what an ordinary (Euclidean) plane is, and even of what three-dimensional space is. Two of our examples are aimed at extending this intuition to projective planes.

Recall that to give a projective plane we must give the points, the lines and the incidences, and then check that the three axioms are satisfied.

1. Euclidean plane plus a line

There are two types of points, *ordinary* and *ideal* points. The ordinary points are the points of the Euclidean plane, and there is one ideal point ("off at infinity somewhere") for each family of parallel lines in the plane.

If the student thinks of the Euclidean plane as the *cartesian* (or *coordinate*) plane, then the ordinary points are pairs (x, y) of real numbers, and, since there is one *gradient* for every family of parallel lines, the ideal points may be thought of as real numbers m (positive or negative) or ∞ .

The lines are just the ordinary Euclidean lines plus one extra line called the *ideal* line. The ordinary lines are represented by equations

$$ax + by + c = 0,$$

where not both a, b are zero. So we may think of the ordinary lines as triples (a, b, c) where not both a, b are zero, and the same line is given by $(\alpha a, \alpha b, \alpha c)$ for any real $\alpha \neq 0$. The ideal line can be included in this scheme as represented by all the triples $(0, 0, c)$ where $c \neq 0$. So we can say that the lines are represented by triples (a, b, c) where not all of a, b, c are zero, and the same line is represented by $(\alpha a, \alpha b, \alpha c)$ for any real $\alpha \neq 0$. These are called *homogeneous line coordinates*.

Now we give the incidences. Ordinary points lie on ordinary lines if they do in the ordinary sense: the point (x, y) is on the ordinary line represented by (a, b, c) when $ax + by + c = 0$. The ideal points all lie on the ideal line and no other points do. Each ordinary line goes through that ideal point which is associated with the family of parallels of which the ordinary line is a member; the ideal point associated with the family of parallels of which the line (a, b, c) is a member is $-\frac{a}{b}$ where this means ∞ when $b = 0$), this being the gradient of the line.

Now we verify the axioms. This may be done using Euclidean geometry or using the algebra.

Axiom 1

Given two distinct points we must see that there is exactly one line which goes through both. If the two points are ordinary this is a postulate of Euclid. Algebraically, if the points are $(x_1, y_1), (x_2, y_2)$ then the line is

$$(y_1 - y_2, x_2 - x_1, x_1y_2 - x_2y_1).$$

If the two points are ideal then the ideal is the only line through both. Euclid's fifth postulate says that there is exactly one line of a given family of parallels which goes through a given point; this exactly tells us that there is exactly one ordinary line through a given ideal point and a given ordinary point. So if one of the points is ordinary and one ideal there is still only one line through them both. (The ideal line does not go through them both since it has no ordinary points on it). Algebraically, given an ordinary point (x_1, y_1) and an ideal point represented by m , the line represented by $(m, -1, y_1 - mx_1)$ is the only line through both.

Axiom 2

This is the axiom which does not hold for the Euclidean plane since parallel lines do not meet. What we have done here is to throw in extra points as points of intersection of parallel lines. Take two distinct ordinary lines. If they meet there is no bother, otherwise they are parallel and they meet in the ideal point corresponding to that family of parallels. Algebraically, the ordinary lines represented by (a, b, c) , (a', b', c') meet at an ordinary point if and only if $ab' - a'b \neq 0$;

but if $ab' - a'b = 0$ then they meet at the ideal point $-\frac{a}{b} = -\frac{a'}{b'}$.

An ordinary line meets the ideal line in a point, namely the ideal point corresponding to the gradient of the ordinary line.

Axiom 3

The four points, $0, \infty, (0, 0), (1, 1)$ are no three collinear, as are the four points $(0, 0), (0, 1), (1, 1), (1, 0)$, etc.

The next example may seem a little more concrete than the last. The reason the last one is not completely concrete is that the mind has the problem as to where to picture this extra ideal line. The same type of problem arose on introducing the square root of minus one; the reason the mind struggles with this is that it wants to put $i = \sqrt{-1}$ on the ordered real number line. It just does not belong there. Nor does the ideal line belong in the Euclidean plane.

There is something, however, which may upset the student at first, and that is that the points are lines and the lines are planes. Remember that in the definition of a projective plane the points and lines could be elements of any set. Yet there is a link-up between these two examples and this will be explained.

2. Concurrent lines in 3D

Suppose 0 is a fixed point in three dimensional Euclidean space. For the algebra, the points of the space are represented by triples of real numbers which we shall write as column vectors $\begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix} = \underline{x}$, and the point 0 may as well be taken as represented by the zero vector $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \underline{0}$.

The points of the projective plane are the lines of the space which pass through the point 0 ; such lines will be called *0-lines*. There is exactly one 0 -line through any point $\underline{x}$ in the space, and the points on the 0 -line through $\underline{x}$ are the ones with coordinates $\alpha \underline{x}$ where α is real. The point 0 is on all the 0 -lines and any other point on an 0 -line completely determines the 0 -line. So we may think of an 0 -line as being represented by a vector $\underline{x}$ where the same 0 -line is represented by the vector $\alpha \underline{x}$ where α is a non-zero real number.

The lines of the projective plane are the planes of the space which pass through the point 0; such planes will be called *0-planes*. The set of points $\underline{x}$ on an 0-plane satisfy an equation

$$u_0 x_0 + u_1 x_1 + u_2 x_2 = 0.$$

So we may think of an 0-plane as being represented by a row vector $(u_0, u_1, u_2) = \underline{u}^T$, and the same 0-plane is represented by the row vector $(\alpha u_0, \alpha u_1, \alpha u_2) = \underline{u}^T$ where α is a non-zero number.

A point of the projective plane is incident with a line of the projective plane when the 0-line is on the 0-plane. If the 0-line is represented by $\underline{x}$ and the 0-plane by $\underline{u}^T$, then the set of points of the 0-line (other than 0) are of the form $\alpha \underline{x}$ ($\alpha \neq 0$) and the condition that this should be on the 0-plane $\underline{u}^T$ is

$$u_0 x_0 + u_1 x_1 + u_2 x_2 = 0;$$

that is, $\underline{u}^T \underline{x} = 0.$

Axiom 1.

Two distinct 0-lines determine a unique 0-plane since any two distinct lines in space which intersect determine a unique plane containing both. The 0-plane determined by the 0-lines represented by $\underline{x}, \underline{y}$ is represented by the row vector $(x_1 y_2 - x_2 y_1, x_2 y_0 - x_0 y_2, x_0 y_1 - x_1 y_0).$

Axiom 2.

Any two planes in space which are not parallel meet in a line. Two 0-planes cannot be parallel since they meet at least at 0. So they meet in an 0-line. The 0-line on both the 0-planes represented by $\underline{u}^T, \underline{v}^T$ is represented by the vector

$$\begin{pmatrix} u_1 v_2 - u_2 v_1 \\ u_2 v_0 - u_0 v_2 \\ u_0 v_1 - u_1 v_0 \end{pmatrix}.$$

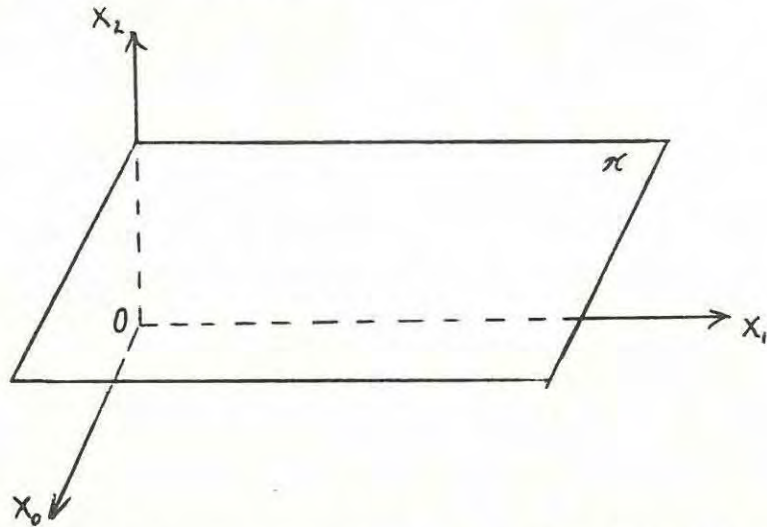
Axiom 3.

The 0-lines through the points $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ have the property that no three are in the one 0-plane.

The connection between these two examples will now be explained.

To relate the two examples, consider the point 0 in three-dimensional space and the Euclidean plane π as any plane not through 0.

For the algebra we may as well take 0 to be the origin and π to be the plane given by the equation $x_2 = 1$.



Every point of π gives rise to a unique 0-line by joining the point to 0. All the 0-lines arise in this way *except* the ones in the plane given by the equation $x_2 = 0$. So there is one line of points in example 2 more than there are points of π . So by adding the ideal line to π as in example 1 we see how to establish a 1-1 correspondence between the points of the two examples under which lines go to lines. Algebraically, the correspondence is as follows:

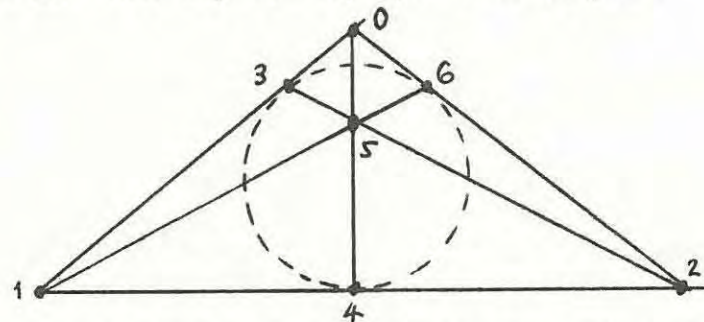
point (x, y) of π	$\longleftrightarrow$	0-line through $\begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$
point m of ideal line	$\longleftrightarrow$	0-line through $\begin{pmatrix} m \\ -1 \\ 0 \end{pmatrix}$
line (a, b, c) of π	$\longleftrightarrow$ a, b not both zero	0-plane (a, b, c)
ideal line	$\longleftrightarrow$ $u_2 \neq 0$	0-plane $(0, 0, u_2)$;
and back the other way:		
0-line through $\begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix}$	$\longleftrightarrow$	point $\frac{x_0}{x_2}, \frac{x_1}{x_2}$ of π if $x_2 \neq 0$ point $-\frac{x_0}{x_1}$ of ideal line $x_2 = 0$
0-plane (u_0, u_1, u_2)	$\longleftrightarrow$	line (u_0, u_1, u_2) of π if not both u_0, u_1 are zero. ideal line if $u_0 = u_1 = 0$.

3. Fano plane

There are seven points $P_0, P_1, P_2, P_3, P_4, P_5, P_6$ and seven lines $\ell_0, \ell_1, \ell_2, \ell_3, \ell_4, \ell_5, \ell_6$ and the incidences are given by the table:

ℓ_0	ℓ_1	ℓ_2	ℓ_3	ℓ_4	ℓ_5	ℓ_6
P_0	P_1	P_2	P_3	P_4	P_5	P_6
P_1	P_2	P_3	P_4	P_5	P_6	P_0
P_3	P_4	P_5	P_6	P_0	P_1	P_2

the points on the line ℓ_i are the three points appearing below ℓ_i in this table. This may be displayed in a diagram:



where P_i is denoted by i and the straight lines together with the dotted line denote the collinear sets of points. From this figure the axioms may easily be checked.

4. Thirteen point plane

The points may be taken as $0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12$ and the collinear sets are the columns in the matrix

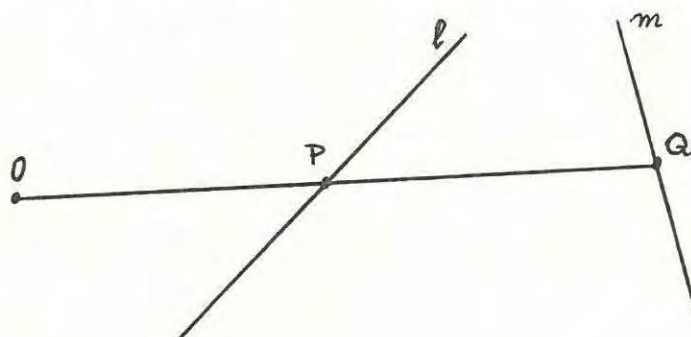
0	1	2	3	4	5	6	7	8	9	10	11	12
1	2	3	4	5	6	7	8	9	10	11	12	0
3	4	5	6	7	8	9	10	11	12	0	1	2
7	8	9	10	11	12	0	1	2	3	4	5	6
9	10	11	12	0	1	2	3	4	5	6	7	8

§4. Perspectivities and projectivities (Read Ch. 4 of Tuller).

In this section we shall suppose we have a projective plane, and so "point" and "line" will relate to this particular projective plane.

Suppose ℓ, m are distinct lines and O is a point not on either ℓ or m . Then there is a 1-1 correspondence set up between the points of ℓ and the points of m , namely:

The point P on ℓ corresponds to the point $Q = (O \vee P) \wedge m$ on m .



The inverse of this transformation is given by:

the point Q on m corresponds to the point $(O \vee Q) \wedge \ell$ on ℓ .

We say that the point P on ℓ is *in perspective* from O with the point Q on m , and write

$$P \stackrel{O}{\wedge} Q.$$

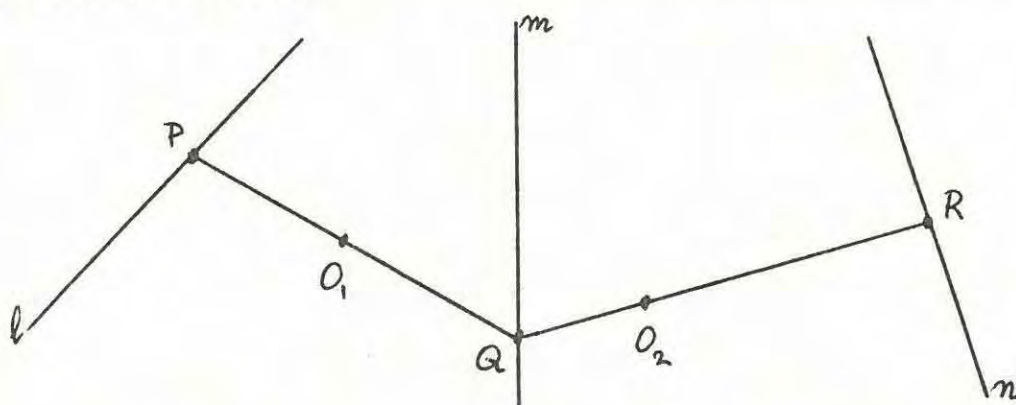
If $P_0 \stackrel{O}{\wedge} Q_0$, $P_1 \stackrel{O}{\wedge} Q_1$, $P_2 \stackrel{O}{\wedge} Q_2$, ... then we write

$$P_0 P_1 P_2 \dots \stackrel{O}{\wedge} Q_0 Q_1 Q_2 \dots$$

This transformation T from the pencil of points on ℓ to the pencil of points on m is called a *perspectivity*, and it is given by

$$T(P) = (O \vee P) \wedge m.$$

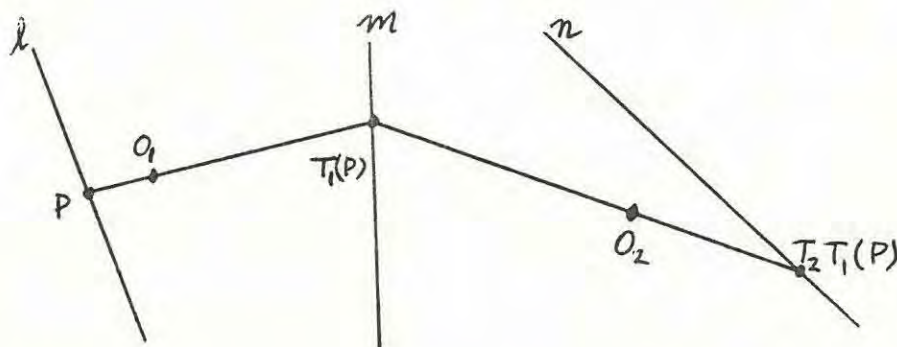
Since perspectivities are transformations they can be composed. Let ℓ, m, n be lines



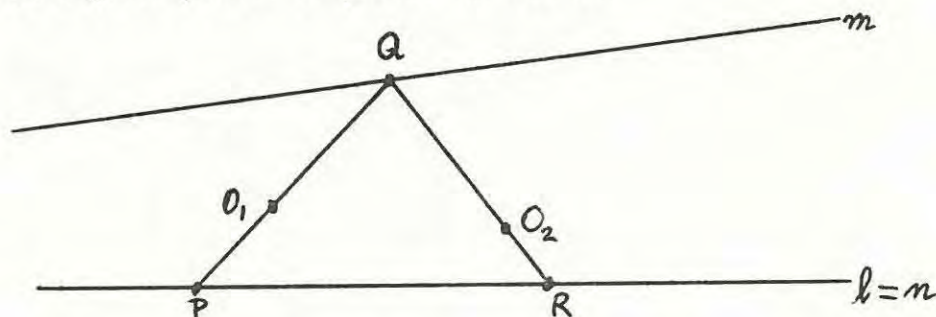
($\ell \neq m$, $m \neq n$) and suppose O_1 is not on ℓ or m and O_2 is not on m or n . Let T_1 be the perspectivity from the pencil of points on ℓ

to the pencil of points on m determined by O , and let T_2 be the perspectivity from the pencil of points of m to the pencil of points on n determined by O_2 . The the *composite* (called "product" by Tuller) transformation $T = T_2T_1$ is given by

$$T(P) = T_2(T_1(P)) = (O_2 \vee ((O_1 \vee P) \wedge m)) \wedge n \text{ for } P \text{ on } \ell.$$



Any finite composite of perspectivities is called a *projectivity*. Although perspectivities were between pencils on distinct lines, projectivities can be from a given pencil of points to itself.



Theorem: *There is a 1-1 correspondence between points of any two lines of a projective plane. In particular, if one line has $k+1$ points on it then they all do.*

Proof: Suppose ℓ, m are distinct lines. By axiom 3 there are four distinct points no three collinear, say P_1, P_2, P_3, P_4 . If one of these is not on either ℓ or m then choose it and call it O . If they are all either on ℓ or m then two must be on ℓ , say P_1, P_2 , and two must be on m , say P_3, P_4 . Then put $O = (P_1 \vee P_3) \wedge (P_2 \vee P_4)$, and O is not on either ℓ or m since no three of the P_i are collinear. So there is always a point O not on either ℓ or m . The perspectivity determined by O sets up the required 1-1 correspondence. //

Theorem: *Given three distinct points A, B, C on a line ℓ and three distinct points A', B', C' on a line m , then there exists a projectivity T from the pencil of points on ℓ to the pencil of points on m ($T: \ell \rightarrow m$)**

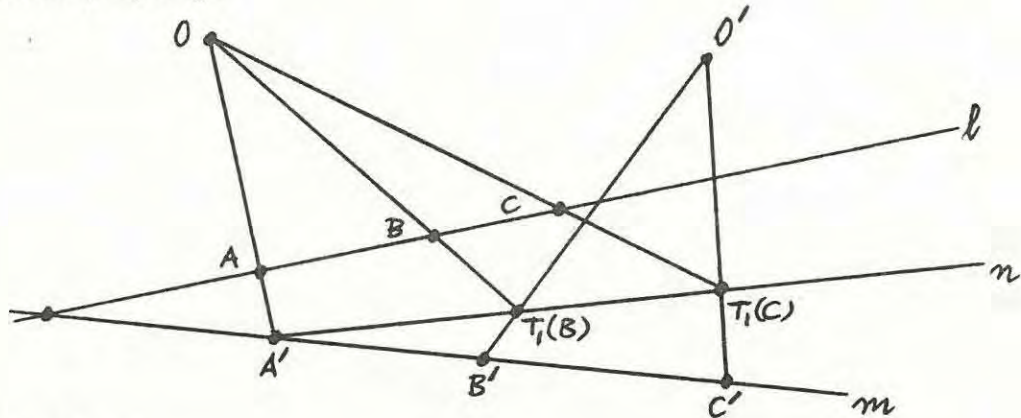
* We often identify a line with the pencil of points on it and think of transformations as functions.

such that $T(A) = A'$, $T(B) = B'$, $T(C) = C'$.

Proof: (a) $\ell \neq m$.

If $A = A'$ ($= \ell \wedge m$) then let T be the perspectivity from the point $(B \vee B') \wedge (C \vee C')$.

If $A \neq A'$, $B \neq B'$, $C \neq C'$ then take O on $A \vee A'$ but not on ℓ or m . There are at least three lines through A' , so let n be a line through A' distinct from m and $A \vee A'$. Let $T_1 : \ell \rightarrow n$ be the perspectivity determined by O ; so $T_1(A) = A'$. Let $O' = (T_1(B) \vee B') \wedge (T_1(C) \vee C')$, and let $T_2 : n \rightarrow m$ be the perspectivity determined by O' . So $T_2(A') = A'$, $T_2(T_1(B)) = B'$, $T_2(T_1(C)) = C'$. Then $T = T_2 T_1$ has the required property.

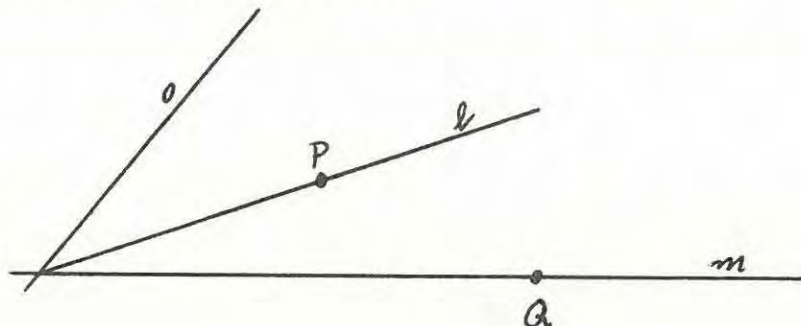


(b) $\ell = m$

Choose any line $\ell' \neq \ell$ and any perspectivity T_0 from ℓ onto ℓ' . By part (a) there is a projectivity T_3 which takes $T_0(A)$, $T_0(B)$, $T_0(C)$ to A' , B' , C' . Then $T = T_3 T_0$ has the required property. //

The dual of "in perspective from a point" is "in perspective from a line". A line o sets up a 1-1 correspondence between the pencil of lines through a point P and the pencil of lines through another distinct point Q (provided o does not go through P or Q); this correspondence is given by:

the line ℓ through P corresponds to the line $m = (o \wedge \ell) \vee Q$ through Q .



We write $\ell \vee \frac{Y}{O} m$. The transformation $T: P \rightarrow Q$ from the pencil of lines through P to the pencil of lines through Q given by $T(\ell) = (o \wedge \ell) \vee Q$ is called a *perspectivity with axis o* . If $\ell_0, \ell_1, \ell_2, \dots$ are lines through P and $m_0, m_1, m_2, \dots$ are lines through Q , and if $\ell_0 \wedge m_0 = A_0, \ell_1 \wedge m_1 = A_1, \ell_2 \wedge m_2 = A_2, \dots$ are collinear, then we write

$$l_0 l_1 l_2 \dots \underset{A_0 A_1 A_2 \dots}{\overset{\vee}{=}} m_0 m_1 m_2 \dots$$

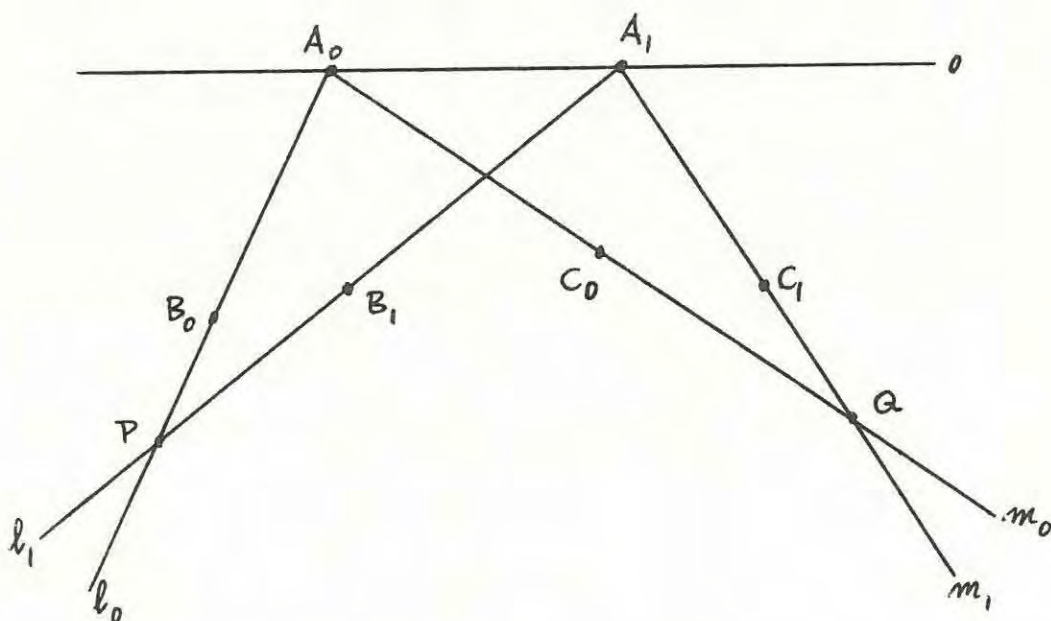
If $B_0, B_1, B_2, \dots$ are points on $\ell_0, \ell_1, \ell_2, \dots$ respectively distinct from P and if $C_0, C_1, C_2, \dots$ are points on $m_0, m_1, m_2, \dots$ respectively distinct from Q , then we write

$$P \vee (B_0 B_1 B_2 \dots) \stackrel{v}{=} Q \vee (C_0 C_1 C_2 \dots)$$

instead of

$$l_0 l_1 l_2 \dots \underset{0}{\overset{\vee}{=}} m_0 m_1 m_2 \dots$$

(Note that $\ell_0 = P \vee B_0$, $\ell_1 = P \vee B_1$, $\ell_2 = P \vee B_2$, ... and $m_0 = Q \vee C_0$, $m_1 = Q \vee C_1$, $m_2 = Q \vee C_2$, ...).



Composites of perspectivities between pencils of lines are projectivities between pencils of lines.

Write out the duals of the last two theorems.

If there exists a projectivity taking collinear points $A, B, C, \dots$ to collinear points $A', B', C', \dots$ then we write

$$A \ B \ C \ \dots \ \overline{\wedge} \ A' \ B' \ C' \ \dots$$

Dually, if there exists a projectivity taking concurrent lines $\ell, m, n, \dots$ to concurrent lines $\ell', m', n', \dots$ then we write

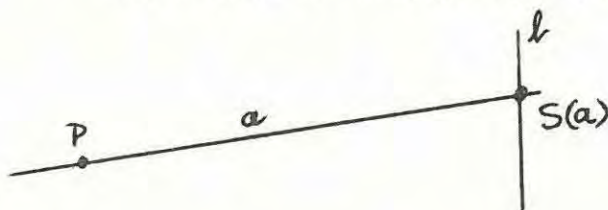
$$\ell m n \dots \vee \ell' m' n' \dots$$

If $\ell, m, n, \dots$ all pass through P and $A, B, C, \dots$ are other points on them respectively, and if $\ell', m', n', \dots$ all pass through P' and $A', B', C', \dots$ are other points on them respectively then we write

$$P \vee (A B C \dots) \vee P' \vee (A' B' C' \dots).$$

There is a link between projectivities for pencils of points and projectivities for pencils of lines. Given any point P and any line ℓ not through P there is a transformation $S : P \rightarrow \ell$ from the pencil of lines through P to the pencil of points on ℓ given by

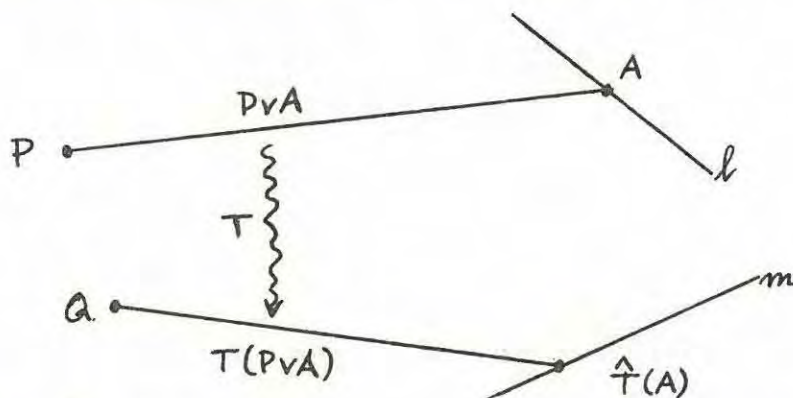
$$\text{for a line } a \text{ through } P, S(a) = a \wedge \ell.$$



The transformation S is called the *section* from P to ℓ .

Suppose the point P is not on the line ℓ and the point Q is not on the line m . A transformation $T : P \rightarrow Q$ determines a transformation $\hat{T} : \ell \rightarrow m$ as follows:

$$\text{for each point } A \text{ on } \ell, \hat{T}(A) = T(P \vee A) \wedge m.$$



The transformation $\hat{T} : \ell \rightarrow m$ may be thought of as the composite of $S^{-1} : \ell \rightarrow P$ and $T : P \rightarrow Q$ and $S : Q \rightarrow m$. We call $\hat{T} : \ell \rightarrow m$ the *transformation induced by* $T : P \rightarrow Q$.

Theorem: Suppose P, Q are points not on lines ℓ, m respectively. If the transformation $T : P \rightarrow Q$ is a projectivity then the transformation

$T : \ell \rightarrow m$ induced by T is a projectivity.

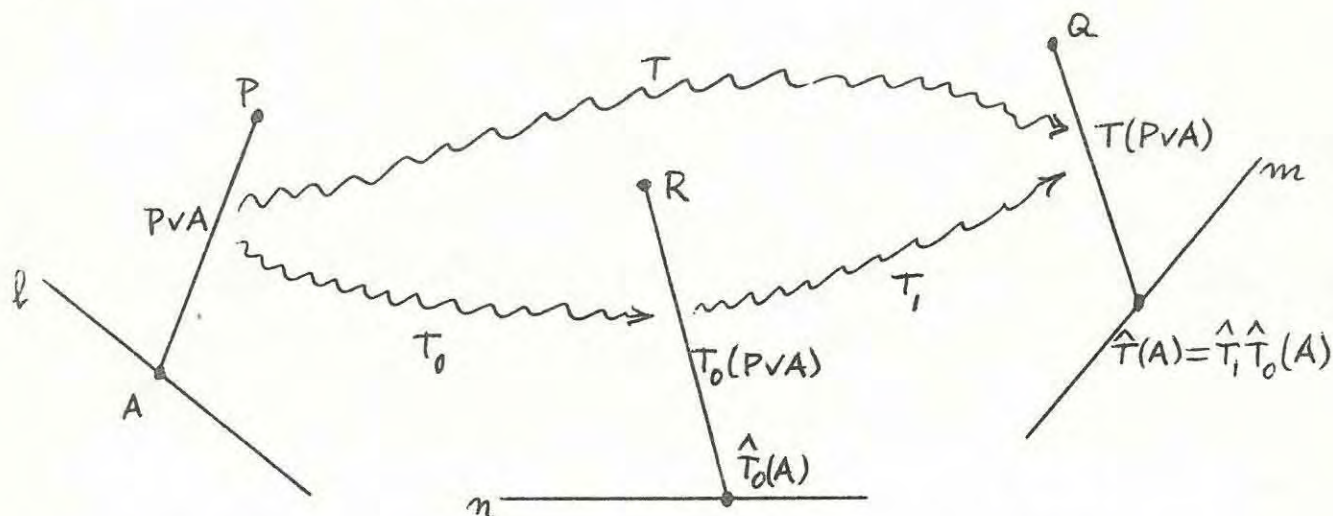
Proof: (a) First we show that the transformation induced by a composite is the composite of the induced transformations. Suppose R is not on n and the transformations $T_0 : P \rightarrow R$, $T_1 : R \rightarrow Q$ induce transformations $\hat{T}_0 : \ell \rightarrow n$, $\hat{T}_1 : n \rightarrow m$, and suppose $T = T_1 T_0 : P \rightarrow Q$.

Then,

$$\begin{aligned}\hat{T}_1(\hat{T}_0(A)) &= \hat{T}_1(T_0(P \vee A) \wedge n) \\ &= T_1(R \vee (T_0(P \vee A) \wedge n)) \wedge m \\ &= T_1(T_0(P \vee A)) \wedge m \text{ since } T_0(P \vee A) \text{ goes through } R, \\ &= \hat{T}(A),\end{aligned}$$

so

$$\hat{T} = \hat{T}_1 \hat{T}_0.$$

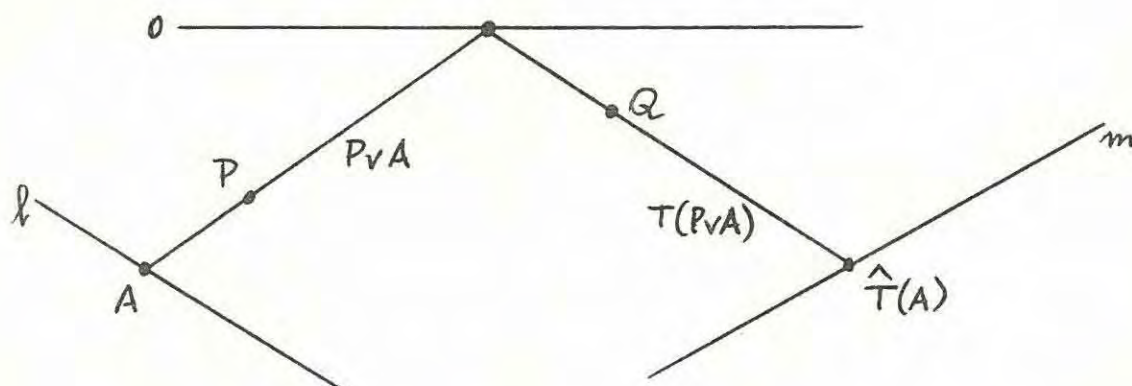


(b) Next we consider the case where $T : P \rightarrow Q$ is a perspectivity. Then there is a line o not through P or Q such that $T(a) = (o \wedge a) \vee Q$ for a through P . So, for A on ℓ ,

$$\begin{aligned}\hat{T}(A) &= T(P \vee A) \wedge m \\ &= ((o \wedge (P \vee A)) \wedge Q) \wedge m\end{aligned}$$

Let $U_0 : \ell \rightarrow o$ be the perspectivity determined by P and $U_1 : o \rightarrow m$ be the perspectivity determined by Q . Then, for A on ℓ ,

$$\begin{aligned}U_1(U_0(A)) &= U_1(o \wedge (P \vee A)) \\ &= m \wedge (Q \vee (o \wedge (P \vee A))) \\ &= \hat{T}(A).\end{aligned}$$



So $\hat{T}$ is the composite of the two perspectivities U_0 and U_1 and hence $\hat{T}$ is a projectivity as required.

(c) Finally we prove the general case. A projectivity $T : P \rightarrow Q$ is a finite composite $T = T_i T_{i-1} \dots T_1 T_0$ of perspectivities T_j . By part (a), $\hat{T} = \hat{T}_i \hat{T}_{i-1} \dots \hat{T}_1 \hat{T}_0$. By part (b), each $\hat{T}_j$ is a projectivity. So $\hat{T}$ is a finite composite of projectivities and so is a projectivity. //

Write the dual theorem and its proof.

Corollary: If $P \vee (A B C \dots) \vee P' \vee (A' B' C' \dots)$ where $A, B, C, \dots$ are collinear and $A', B', C', \dots$ are collinear then $A B C \dots \wedge A' B' C' \dots$.

Proof: Let $T : P \rightarrow P'$ be a projectivity which takes $P \vee A, P \vee B, P \vee C, \dots$ to $P' \vee A', P' \vee B', P' \vee C', \dots$ respectively. Let ℓ be the line $A \vee B \vee C \vee \dots$ and let ℓ' be the line $A' \vee B' \vee C' \vee \dots$. Let $\hat{T} : \ell \rightarrow \ell'$ be the transformation induced by T . By the last theorem $\hat{T} : \ell \rightarrow \ell'$ is a projectivity. Moreover,

$$\begin{aligned} \hat{T}(A) &= T(P \vee A) \wedge \ell' \\ &= (P' \vee A) \wedge \ell' \\ &= A', \end{aligned}$$

and similarly $\hat{T}(B) = B', \hat{T}(C) = C', \dots$. So there is a projectivity which takes $A, B, C, \dots$ to $A', B', C', \dots$. //

§5. Desarguesian and Pappusian planes

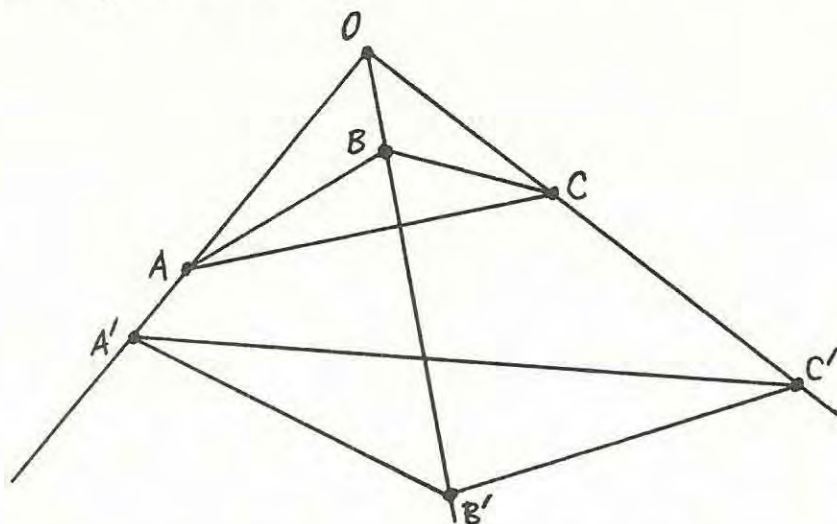
A *point triangle* (strictly, an *ordered point triangle*) in a projective plane is simply a set of three distinct points (called *vertices*) taken in some order. The point triangle with vertices A, B, C taken in the order first A , then B , then C is denoted by $\Delta A B C$.

The dual of a point triangle is a *line triangle*. A line triangle is a set of three distinct lines (called *sides*) taken in some order. The line triangle with sides a, b, c taken in that order is denoted by $\Delta a b c$.

A point triangle $\Delta A B C$ gives rise to a line triangle $\Delta a b c$ where $a = B \vee C$, $b = C \vee A$, $c = A \vee B$; the lines a, b, c are also called the sides of the point triangle. Similarly (dually), a line triangle gives rise to a point triangle; in particular, the line triangle $\Delta a b c$ gives rise to the point triangle $\Delta A B C$ again. So we freely identify $\Delta A B C$ and $\Delta a b c$ and just call it a *triangle*; the triangle is determined by either its set of vertices or by its set of sides. With A, B, C, a, b, c as above, a is called *the side opposite the vertex A* , B is *the vertex opposite the side b* , etc.

Two triangles $\Delta A B C, \Delta A' B' C'$ are *in perspective from a point* O if the lines $A \vee A', B \vee B', C \vee C'$ are concurrent meeting at O . (Notice how the ordering is important: if $\Delta A B C, \Delta A' B' C'$ are in perspective from a point, the triangles $\Delta A B C, \Delta A' C' B'$ need not be).

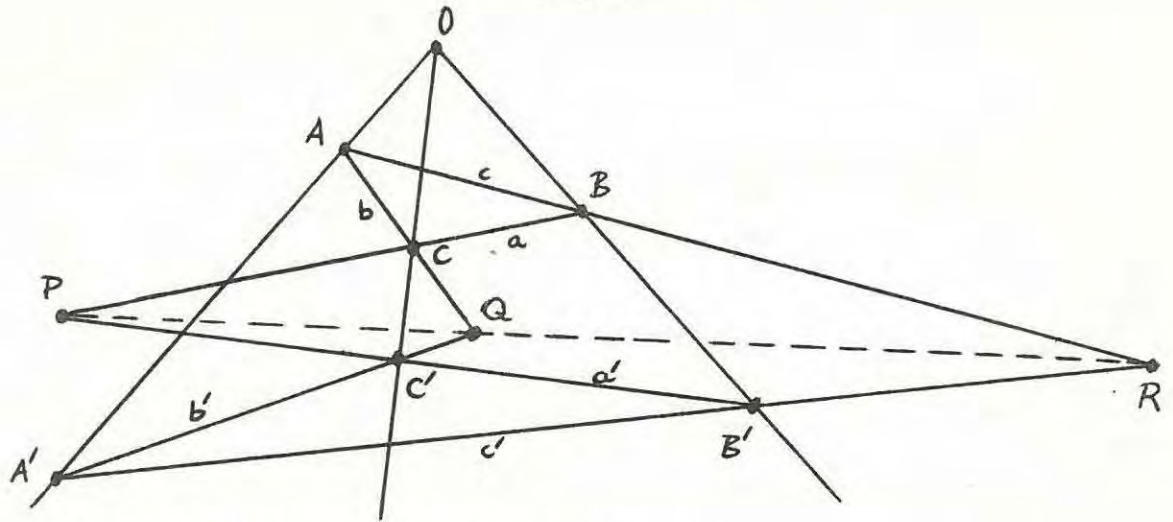
Write $\Delta A B C \overset{O}{\sim} \Delta A' B' C'$.



Dually, two triangles $\Delta a b c, \Delta a' b' c'$ are *in perspective from a line* o if the points $a \wedge a', b \wedge b', c \wedge c'$ are collinear lying on o . Write $\Delta a b c \overset{o}{\sim} \Delta a' b' c'$.

A projective plane is called a *Desarguesian plane* when it satisfies the following axiom:

Desargues's Axiom: Any two triangles which are in perspective from a point are in perspective from a line.



This may be expressed symbolically as:

$$\Delta ABC \stackrel{O}{\wedge} \Delta A'B'C' \implies (B \vee C) \wedge (B' \vee C'), (C \vee A) \wedge (C' \vee A'), (A \vee C) \wedge (A' \vee B') \text{ are collinear.}$$

The reader is requested to draw the diagram for himself to see that "it works" in the Euclidean plane.

Theorem: In a Desarguesian plane, two triangles are in perspective from a point if and only if they are in perspective from a line.

Proof: Suppose $\Delta abc \stackrel{O}{\wedge} \Delta a'b'c'$ and let $A = b \wedge c$, $B = c \wedge a$, $C = a \wedge b$, $A' = b' \wedge c'$, $B' = c' \wedge a'$, $C' = a' \wedge b'$, $P = a \wedge a'$, $Q = b \wedge b'$, $R = c \wedge c'$. Then $\Delta B P B' \stackrel{R}{\wedge} \Delta A Q A'$. So, by Desargues' axiom, $(P \vee B') \wedge (Q \vee A')$, $(B' \vee B) \wedge (A' \vee A)$, $(B \vee P) \wedge (A \vee Q)$ are collinear. That is, C' , $(B' \vee B) \wedge (A' \vee A)$, C are collinear. This means $(B' \vee B) \wedge (A' \vee A)$ is on the line $C' \vee C$. So the lines $A' \vee A$, $B' \vee B$, $C' \vee C$ are concurrent; say they meet at O .

So $\Delta ABC \stackrel{O}{\wedge} \Delta A'B'C'$. So triangles Δabc and $\Delta a'b'c'$ are in perspective from O . This proves that "if" two triangles are in perspective from a line then they are in perspective from a point. This is "only if" by Desargues axiom. //

This theorem shows that the Desargues' axiom implies its own converse. Strangely, the converse of Desargues' axiom is also the dual:

$$\Delta abc \stackrel{O}{\vee} \Delta a'b'c' \implies (b \wedge c) \vee (a' \wedge c'), (c \wedge a) \vee (c' \wedge a), (a \wedge b) \vee (a' \wedge b') \text{ are concurrent.}$$

This may also be written in terms of points. Let ΔABC , $\Delta A'B'C'$ be the triangles Δabc , $\Delta a'b'c'$ and let $a \wedge a' = P$, $b \wedge b' = Q$, $c \wedge c' = R$.

If $\Delta abc \stackrel{O}{\vee} \Delta a'b'c'$ then $o = P \vee Q \vee R$. We write $\Delta ABC \stackrel{PQR}{\vee} \Delta A'B'C'$

when $(B \vee C) \wedge (B' \vee C') = P$, $(C \vee A) \wedge (C' \vee A') = Q$, $(A \vee B) \wedge (A' \vee B') = R$ and P, Q, R are collinear. So the converse of Desargues' axiom may be expressed as:

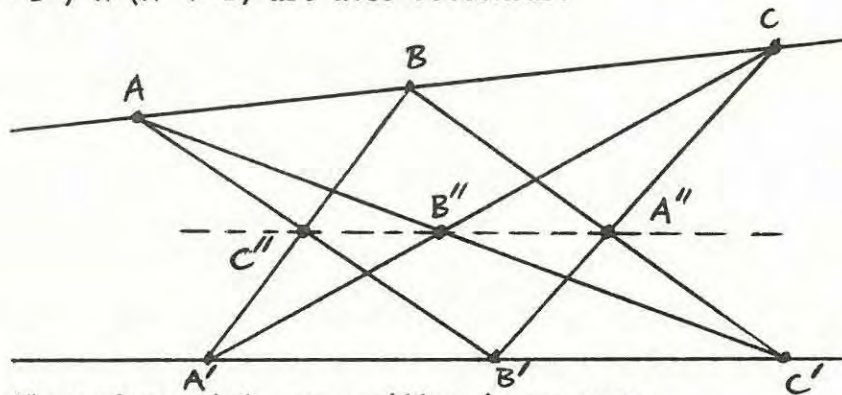
$$\begin{array}{c} \Delta A B C \\ \vee \\ P Q R \end{array} = \Delta A' B' C' \implies A \vee A', B \vee B', C \vee C' \text{ are concurrent.}$$

We have also shown that the duality principle also holds for Desarguesian planes since the duals of all the axioms hold:

if a statement about all Desarguesian planes is true, then so is the dual statement.

A projective plane is called a *Pappusian plane* when it satisfies the following axiom:

Pappus's Axiom: If A, B, C and A', B', C' are two sets of three collinear points then the three points $A'' = (B \vee C') \wedge (B' \vee C)$, $B'' = (C \vee A') \wedge (C' \vee A)$, $C'' = (A \vee B') \wedge (A' \vee B)$ are also collinear.



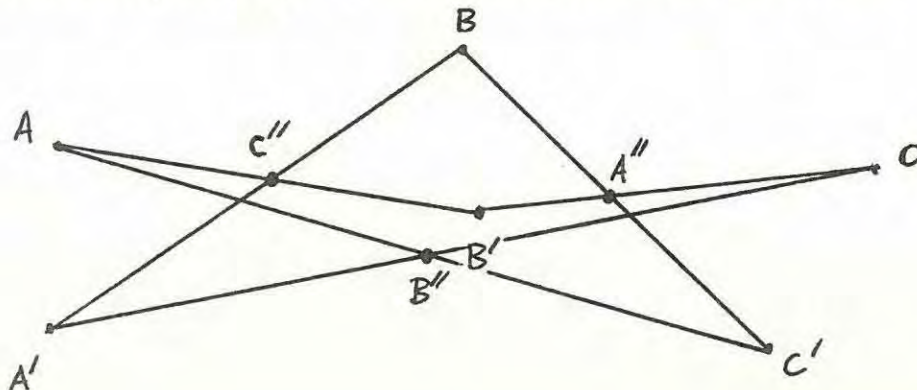
When nine points are written in an array

$$\left\| \begin{array}{ccc} A & B & C \\ A' & B' & C' \\ A'' & B'' & C'' \end{array} \right\|$$

it will mean that

$$A'' = (B \vee C') \wedge (B' \vee C), B'' = (A \vee C') \wedge (A' \vee C), C'' = (A \vee B') \wedge (A' \vee B).$$

In the language of determinants, each point in the last row is equal to its corresponding "minor" (where multiplication is replaced by $\vee$, and subtraction is replaced by $\wedge$). Note that this is then true for all the nine



points, not just those in the last row. For example,

$$C = (A' \vee B'') \wedge (A'' \vee B').$$

So interchanging rows and columns has no affect on the property.

Pappus's axiom becomes:

$$\text{given } \begin{vmatrix} A & B & C \\ A' & B' & C' \\ A'' & B'' & C'' \end{vmatrix}, \quad \text{if two rows are collinear}$$

then so is the third.

Dually, an array

$$\begin{vmatrix} a & b & c \\ a' & b' & c' \\ a'' & b'' & c'' \end{vmatrix}$$

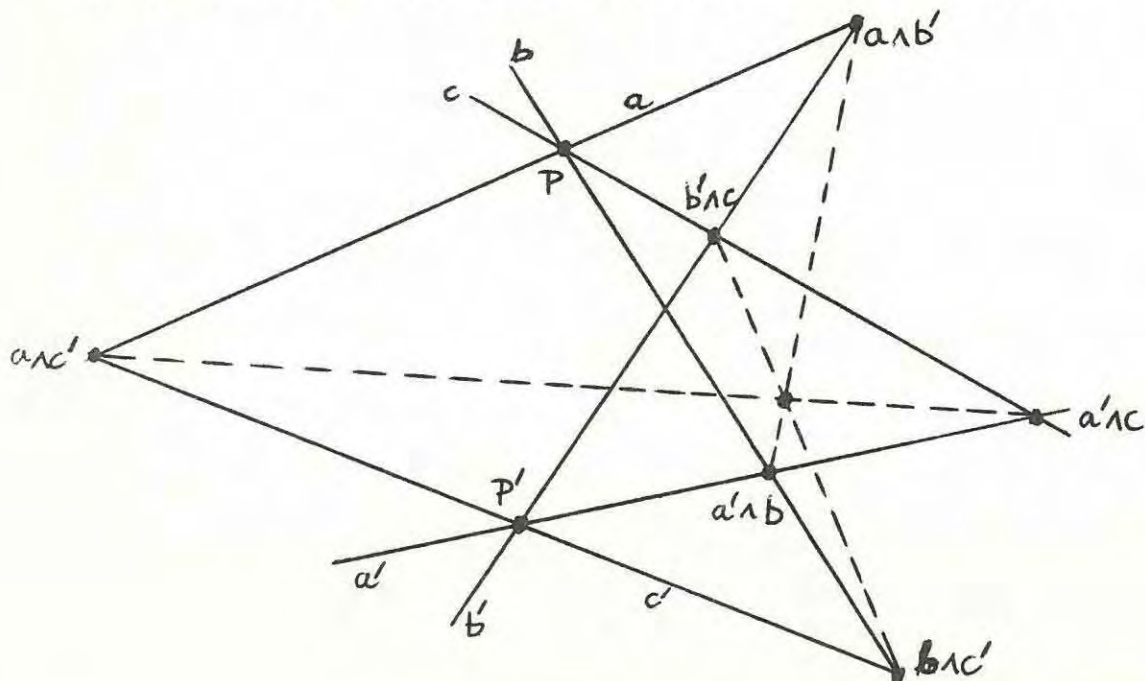
of nine lines means

$$a'' = (b \vee c') \wedge (b' \vee c), \quad b'' = (a \vee c') \wedge (a' \vee c), \quad c'' = (a \vee b') \wedge (a' \vee b).$$

Theorem: In a Pappusian plane, given $\begin{vmatrix} a & b & c \\ a' & b' & c' \\ a'' & b'' & c'' \end{vmatrix}$ if two rows are

concurrent then so is the third.

Proof: Suppose the first two rows are concurrent without loss of generality, and let $P = a \wedge b \wedge c$, $P' = a' \wedge b' \wedge c'$ and $P'' = a'' \wedge c''$.



Then the first two rows of

$$\begin{vmatrix} P & a' \wedge b & b \wedge c' \\ P' & b' \wedge c & a \wedge b' \\ P'' & a \wedge c' & a' \wedge c \end{vmatrix}$$

are collinear, so by Pappus's axiom, the last row is collinear. So $a'' \wedge c''$, $a \wedge c'$, $a' \wedge c$ are collinear; so $b'' = (a \wedge c') \vee (a' \wedge c)$ goes through $a'' \wedge c''$; so a'' , b'' , c'' are concurrent. //

Thus we have the principle of duality for Pappusian planes:

if a statement about all Pappusian planes is true, then so is the dual statement.

Theorem: *Pappusian planes are Desarguesian* (G. Hessenberg, Mathematische Annalen, 61 (1905) pp. 161-172).

Proof: Suppose $\triangle ABC$ $\overset{O}{\sim} \triangle A'B'C'$, and let $P = (B \vee C) \wedge (B' \vee C')$, $Q = (C \vee A) \wedge (C' \vee A')$, $R = (A \vee B) \wedge (A' \vee B')$.

Also put

$$\begin{aligned} S &= (A \vee C) \wedge (B' \vee C'), & T &= (A \vee B') \wedge (O \vee C), \\ U &= (A \vee B) \wedge (O \vee S), & V &= (A' \vee B') \wedge (O \vee S). \end{aligned}$$

The first two rows of the array

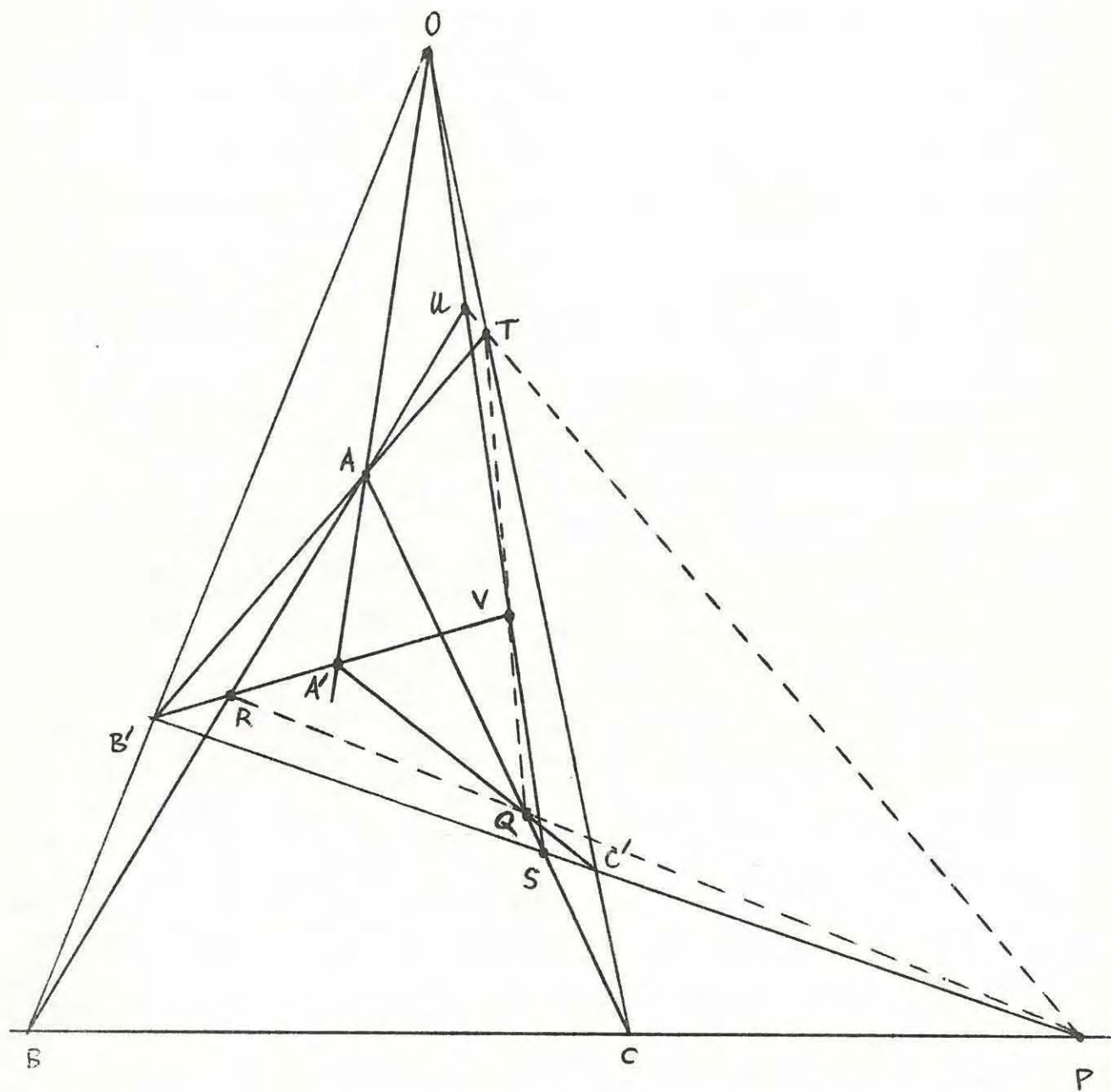
$$\begin{vmatrix} O & B & B' \\ A & S & C \\ P & T & U \end{vmatrix}$$

are collinear, so P, T, U are collinear. The first two rows of the array

$$\begin{vmatrix} O & A & A' \\ B' & C' & S \\ Q & V & T \end{vmatrix}$$

are collinear, so Q, V, T are collinear. The first two rows of the array

$$\begin{vmatrix} A & B' & T \\ V & U & S \\ P & Q & R \end{vmatrix}$$



are collinear, so P, Q, R are collinear as required by Desargues's axiom. //

A projective geometry is *finite* when it has a finite number of points and lines. It is sufficient to know that one line has a finite number of points. The proof of the following result is difficult and will be omitted.

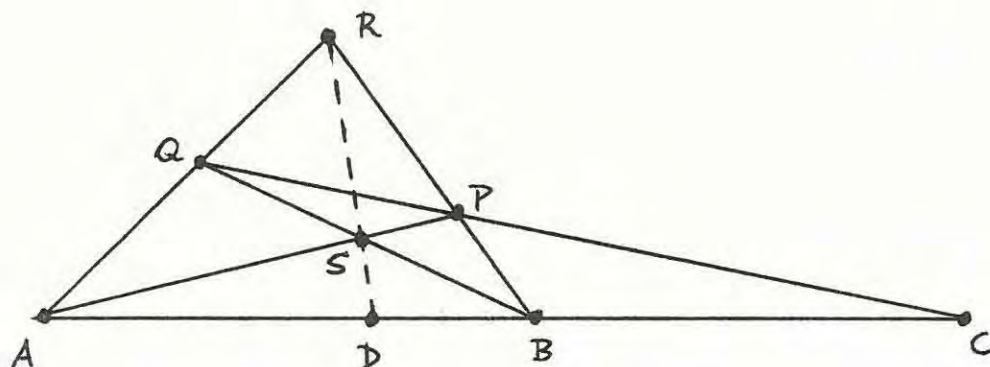
Theorem: *Every finite Desarguesian plane is Pappusian. //*

So for finite planes, Pappus's axiom and Desargues's axiom are equivalent. However, there are infinite planes which satisfy Desargues's axiom but not Pappus's.

§6. Harmonic conjugates.

For this section we suppose we have a Desarguesian plane.

Suppose A, B are given distinct points. To any point C on the line $A \vee B$ we construct another point D as follows. Select any point R not on $A \vee B$, and any point P on $R \vee B$. Let $Q = (P \vee C) \wedge (R \vee A)$, and let $S = (A \vee P) \wedge (B \vee Q)$. Then $D = (R \vee S) \wedge (A \vee B)$.



Then Suppose a different choice R', P' is made for R, P obtaining Q', S' .

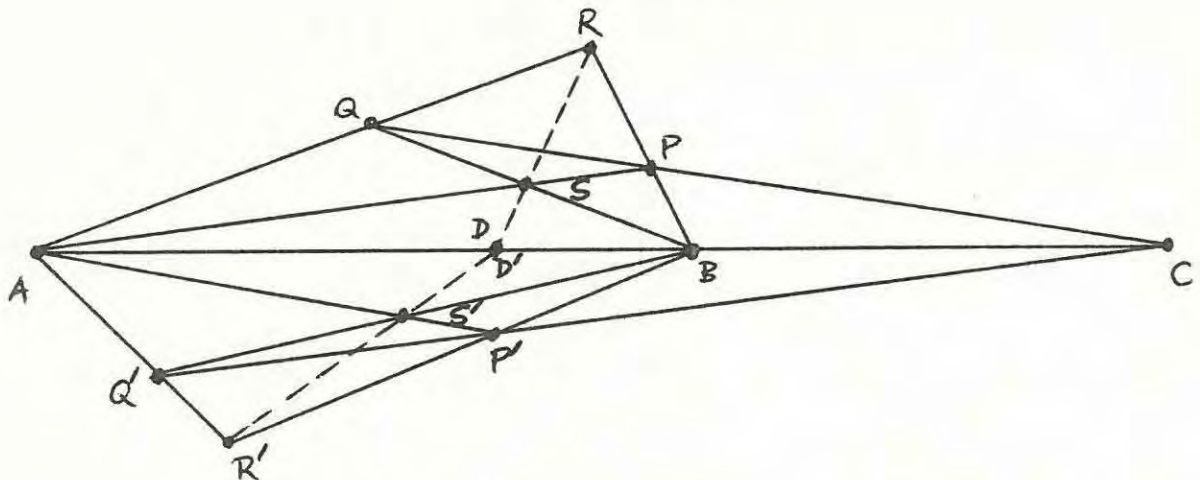
$$\triangle PQR \stackrel{\vee}{\sim} \triangle P'Q'R' \implies P \vee P', Q \vee Q', R \vee R' \text{ concurrent,}$$

$$\triangle PQS \stackrel{\vee}{\sim} \triangle P'Q'S' \implies P \vee P', Q \vee Q', S \vee S' \text{ concurrent.}$$

So $P \vee P', Q \vee Q', R \vee R', S \vee S'$ are concurrent meeting at O say. Then

$$\triangle RSP \stackrel{Q}{\sim} \triangle R'S'P' \implies A, (R \vee S) \wedge (R' \vee S'), B \text{ collinear.}$$

So $R \vee S, R' \vee S$ meet $A \vee B$ at the same point $D = D'$.



Thus we have a process for assigning a unique point D to any point C on the line $A \vee B$. When A, B, C, D are related in this way we say that D is the harmonic conjugate of C with respect to A and B , and we write

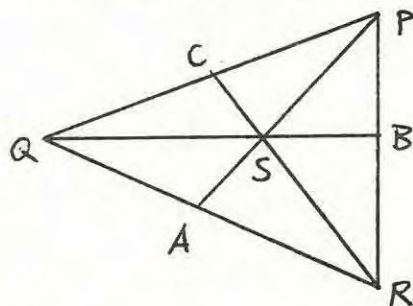
$$A B \underline{H} C D.$$

Let us examine this construction in a little more detail so harmonic conjugates will be recognized in any diagram.

A *complete quadrangle* is a set of four points P, Q, R, S and these points are called *vertices*. The lines $Q \vee R, P \vee S, R \vee P, P \vee Q, R \vee S$ are called *sides*. These sides come in *opposite pairs*:

$$\begin{aligned} Q \vee R, & P \vee S \\ R \vee P, & Q \vee S \\ P \vee Q, & R \vee S. \end{aligned}$$

The intersections of opposite sides are called *diagonal points*.



From the harmonic construction it can be seen that A, B are diagonal points of the complete quadrangle P, Q, R, S and the other two sides of the complete quadrangle, not through A or B , pass through C, D .

So, for four collinear points A, B, C, D , the relation $A B \underline{H} C D$ is satisfied when there exists a complete quadrangle such that two opposite sides pass through A , two other opposite sides pass through B , and the two

remaining sides pass through C and D, respectively,

By symmetry then, $AB \underline{H} CD$ implies $BA \underline{H} CD$, $AB \underline{H} DC$ and $BA \underline{H} DC$.

Theorem: $AB \underline{H} CD$ implies $CD \underline{H} AB$.

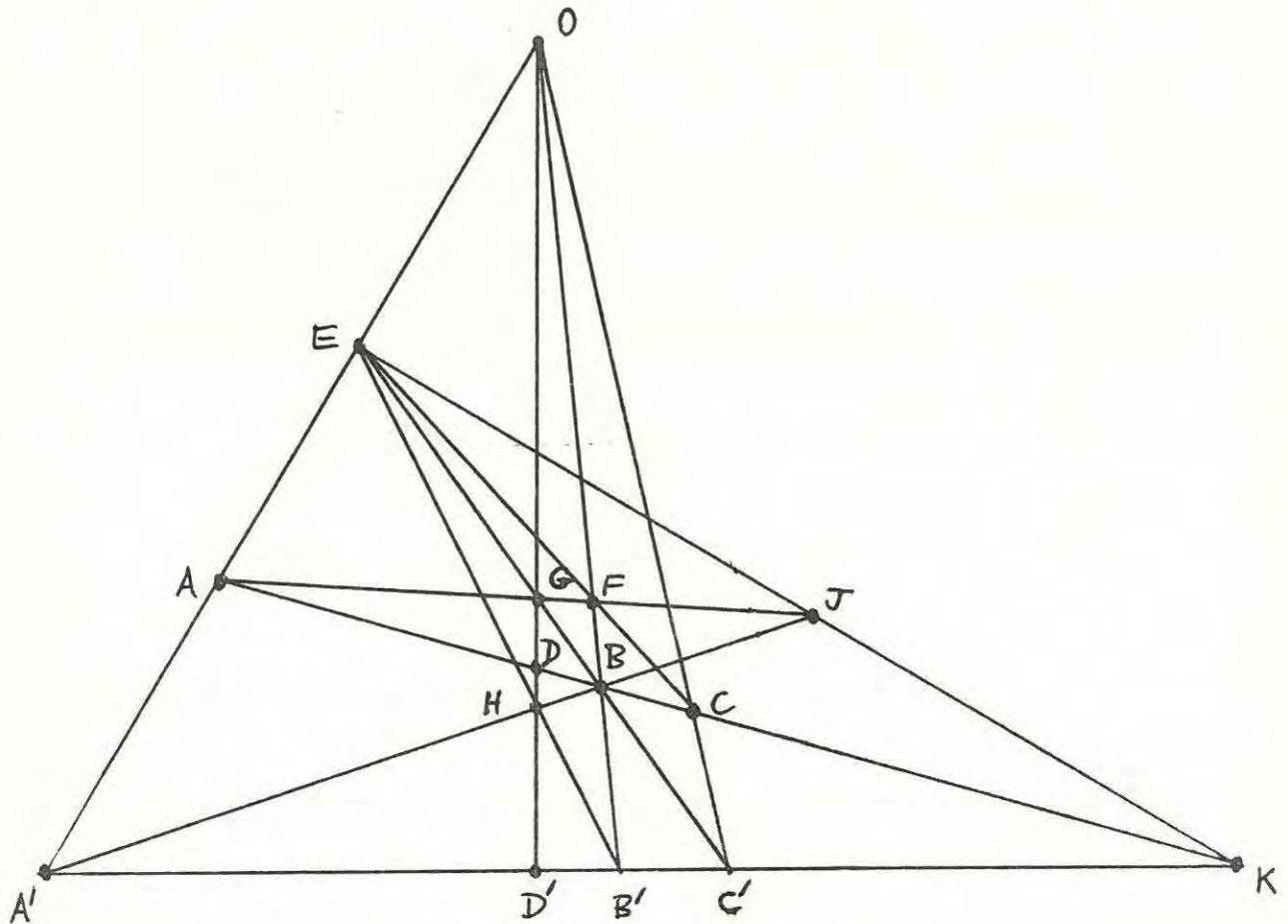
Proof: Suppose $AB \underline{H} CD$ using the quadrangle P, Q, R, S. Let $E = (D \vee P) \wedge (R \vee C)$, $O = (P \vee Q) \wedge (R \vee S)$. Then

$$\Delta QOS \stackrel{V}{\equiv} \Delta REP \implies Q \vee R, S \vee P, O \vee E \text{ concurrent.}$$

So $O \vee E$ passes through A. Then the quadrangle O P E R gives $CD \underline{H} AB$.

Theorem: The harmonic relation is invariant under a projectivity.

Proof: It suffices to show invariance under a perspectivity.



Suppose $ABC \stackrel{O}{\wedge} A'B'C'$ and $(A \vee B) \wedge (A' \vee B') = K$.

Construct $E = (C' \vee B) \wedge (A \vee A')$, $F = (E \vee C) \wedge (O \vee B')$, $G = (A \vee F) \wedge (E \vee B)$,

$H = (E \vee B') \wedge (A' \vee B)$ and $J = (A \vee F) \wedge (A' \vee B)$. By Desargues's axiom,

Then
$$\begin{array}{c} \Delta A F C \\ \parallel \\ \Delta A' B' C' \end{array} \xRightarrow{O} E, K, J \text{ collinear.}$$

$$\begin{array}{c} \Delta J A A' \\ \parallel \\ \Delta E B B' \end{array} \xRightarrow{K} O, H, G \text{ collinear.}$$

Now $D = (O \vee G) \wedge (A \vee B)$ is the point with the property $A B \parallel C D$. Define D' by $A B C D \parallel A' B' C' D'$. Then $D' = (O \vee H) \wedge (A' \vee B')$, so $A' B' \parallel C' D'$. //

Corollary: If $A B \parallel C D$ and $A' B' \parallel C' D'$ then there exists a projectivity of the line $A \vee B$ onto the line $A' \vee B'$ taking A, B, C, D to A', B', C', D' respectively.

Proof: By an earlier theorem there is a projectivity T taking A, B, C to A', B', C' respectively, and by the last theorem, this also takes D to D' . //

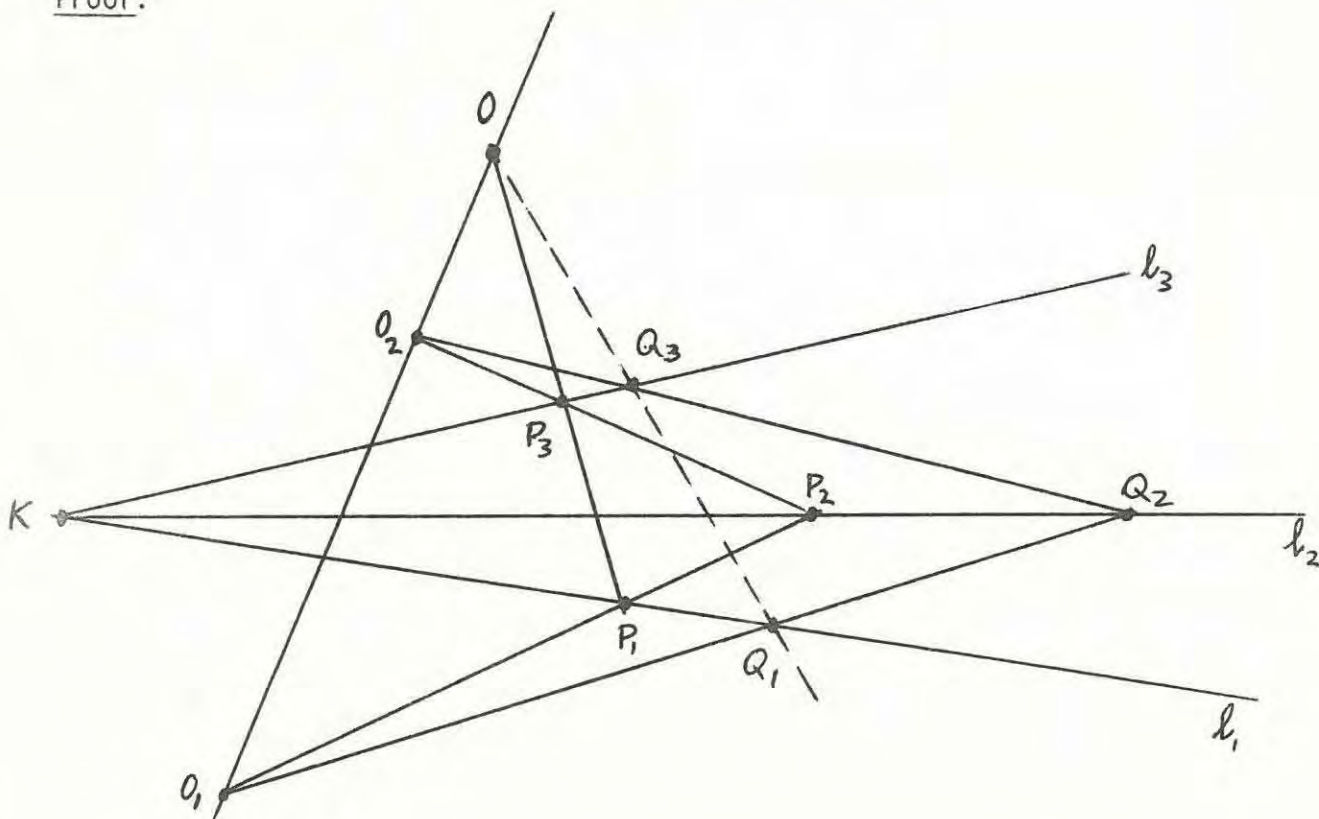
§7. The fundamental theorem of projective geometry

The purpose of this section is to examine projectivities in a Pappusian plane in more detail. A projectivity was defined to be a composite of a finite number of perspectivities; we shall show that any projectivity between distinct lines which is not a perspectivity can be obtained as the composite of just two perspectivities (for projectivities on the same line, three perspectivities may be needed). From this will follow the fundamental theorem of projective geometry which states that a projectivity is uniquely determined when three distinct points of one line and the corresponding three points of the other are given.

In general, the composite of two perspectivities is not a perspectivity; however:

Theorem: In a Desarguesian plane, if $T_1 : l_1 \rightarrow l_2$, $T_2 : l_2 \rightarrow l_3$ are perspectivities and l_1, l_2, l_3 are concurrent, then the composite $T_2 T_1 : l_1 \rightarrow l_3$ is a perspectivity and the centres of $T_1, T_2, T_2 T_1$ are collinear.

Proof:



Let $K = l_1 \cap l_2 \cap l_3$, and let O_1, O_2 be the centres of T_1, T_2 . Choose any point P_1 on l_1 distinct from K and not on $O_1 \vee O_2$. Let $T_1(P_1) = P_2$, $T_2(P_2) = P_3$ and $O = (P_1 \vee P_3) \cap (O_1 \vee O_2)$. Let $T : l_1 \rightarrow l_3$ be the perspectivity with centre O . We claim that $T_2 T_1 = T$. To prove this, take Q_1 on l_1 . Let $Q_2 = T_1(Q_1)$, $Q_3 = T_2(Q_2)$. Then

$$\triangle P_3 P_1 K \stackrel{P_2}{=} \triangle O_2 O_1 Q_2 \implies Q_1, Q_3, O \text{ collinear.}$$

So $T(Q_1) = Q_3 = T_2 T_1(Q_1)$ for all points Q_1 . So $T = T_2 T_1$, a perspectivity with centre O on $O_1 \vee O_2$. //

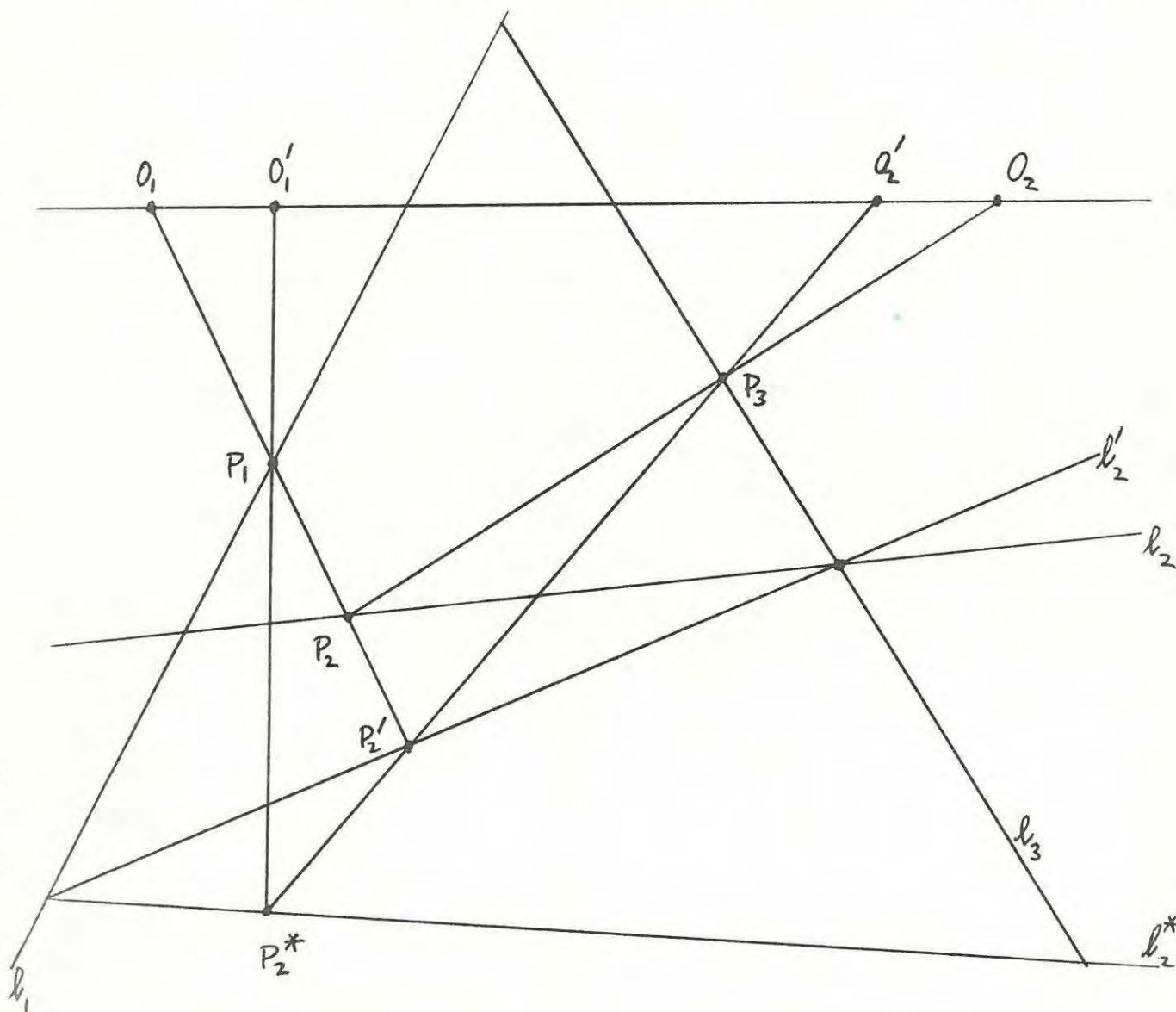
Theorem: In a Desarguesian plane, suppose $T_1 : l_1 \rightarrow l_2$, $T_2 : l_2 \rightarrow l_3$ are perspectivities with centres O_1, O_2 and suppose l_2^* is any line which does not pass through $l_1 \cap l_3$ and which does not pass through two corresponding points of the projectivity $T_2 T_1 : l_1 \rightarrow l_3$. Then there exist two points O_1', O_2' on the line $O_1 \vee O_2$ such that, if $T_1' : l_1 \rightarrow l_2^*$, $T_2' : l_2^* \rightarrow l_3$ denote the projectivities with centres O_1', O_2' then $T_2' T_1' = T_2 T_1$.

(In other words, for a composite of two perspectivities the intermediary line may be replaced by almost any line you like).

Proof: The idea behind this proof is to consider the line l_2' joining $l_1 \cap l_2^*$

and $\ell_2 \wedge \ell_3$, and then to apply the last theorem on "concurrent" perspectivities twice.

Let $S : \ell_2 \rightarrow \ell_2'$ be the perspectivity with centre O_1 . Then $T_2 S^{-1} : \ell_2' \rightarrow \ell_3$ is a composite of two perspectivities $S^{-1} : \ell_2' \rightarrow \ell_2$ and $T_2 : \ell_2 \rightarrow \ell_3$ and so is a perspectivity with centre O_2' say on $O_1 \vee O_2$.



Let $U : \ell_2' \rightarrow \ell_2^*$ denote the perspectivity with centre O_2' . Then $U(ST_1) : \ell_1 \rightarrow \ell_2^*$ is a composite of two perspectivities $ST_1 : \ell_1 \rightarrow \ell_2'$ (centre O_1) and $U : \ell_2' \rightarrow \ell_2^*$ and so is a perspectivity with centre O_1' say on $O_1 \vee O_1'$. Let $T_1' = UST_1 : \ell_1 \rightarrow \ell_2^*$ and let $T_2' = T_2 S^{-1}U^{-1} : \ell_2^* \rightarrow \ell_3$.

Then $T_1' : l_1 \rightarrow l_2^*$ is a perspectivity with centre O_1' , $T_2' : l_2^* \rightarrow l_3$ is a perspectivity with centre O_2' (since U^{-1} and T_2S^{-1} both have centre O_2'), and $T_2'T_1' = T_2S^{-1}U^{-1}UST_1 = T_2T_1$. Also O_1, O_2, O_1', O_2' are collinear. //

(The student should observe why the restrictions on the line l_2^* are needed).

Theorem: Any projectivity between two distinct lines in a Desarguesian plane is the composite of two perspectivities.

Proof: It suffices to show that any sequence of three perspectivities can always be replaced by a sequence of two perspectivities.

Suppose $T_1 : l_1 \rightarrow l_2, T_2 : l_2 \rightarrow l_3, T_3 : l_3 \rightarrow l_4$ are perspectivities. If l_1, l_2, l_3 are concurrent then T_2T_1 is a perspectivity, so $T_3T_2T_1$ is a composite of two perspectivities and we are done. So we may suppose l_1, l_2, l_3 are not concurrent. Similarly for l_2, l_3, l_4 .

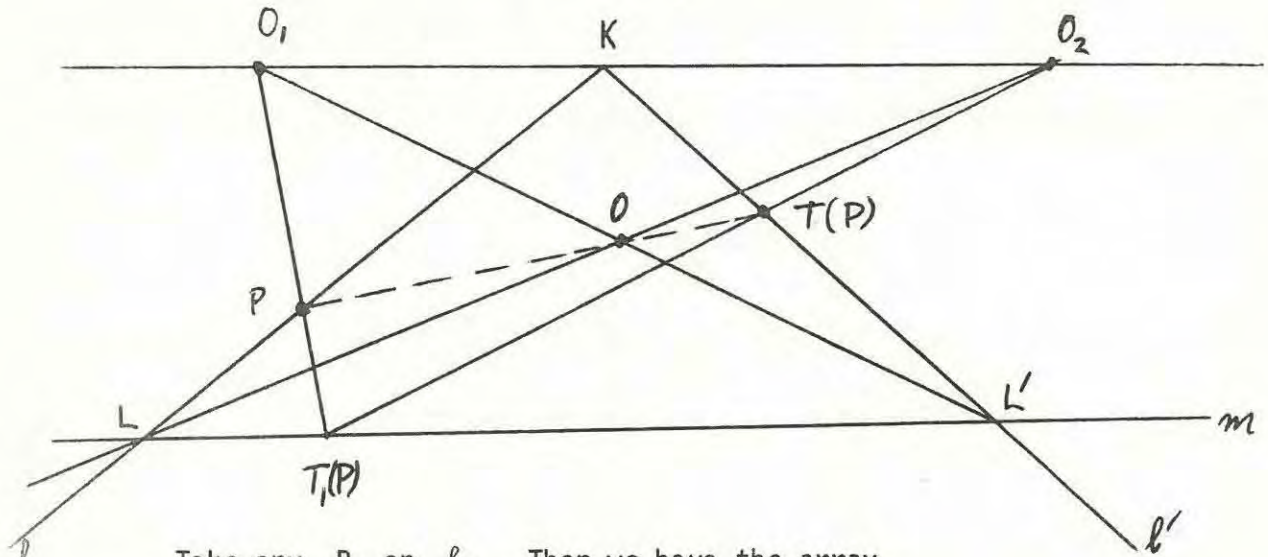
Suppose l_1, l_3, l_4 are not concurrent. Let l_2^* be any line through $l_3 \wedge l_4$ and such that $l_2^* \wedge l_1$ and $l_2^* \wedge l_3$ do not correspond under T_2T_1 . Then l_2^* does not pass through $l_1 \wedge l_3$ since $l_1 \wedge l_3 \neq l_3 \wedge l_4$. By the last theorem there are perspectivities $T_1' : l_1 \rightarrow l_2^*, T_2' : l_2^* \rightarrow l_3$ such that $T_2'T_1' = T_2T_1$. Then $T_3T_2T_1 = T_3T_2'T_1'$ and T_3T_2' is a perspectivity since l_2^*, l_3, l_4 are concurrent. So $T_3T_2T_1$ is a composite of two perspectivities.

Suppose l_1, l_3, l_4 are concurrent. Then l_1, l_2, l_4 are not concurrent (or else l_1, l_2, l_3 would be, contrary to assumption). So, by a similar argument replacing l_3 by a line l_3^* through $l_1 \wedge l_2$ and obtaining perspectivities $\bar{T}_2 : l_2 \rightarrow l_3^*, \bar{T}_3 : l_3^* \rightarrow l_4$ such that $\bar{T}_3\bar{T}_2 = T_3T_2$, again $T_3T_2T_1 = \bar{T}_3\bar{T}_2T_1$ is a composite of two perspectivities. //

Theorem: In a Pappusian plane, a projectivity between two distinct lines is a perspectivity if and only if their point of intersection is a self-corresponding point.

Proof: If $T : l \rightarrow l'$ is a perspectivity with centre O then $T(P) = (O \vee P) \wedge l'$ for P on l . So, if $P = l \wedge l'$, $T(P) = P$. This proves "only if".

Suppose $T : l \rightarrow l'$ is a projectivity and $T(K) = K$ where $K = l \wedge l'$. Since l, l' are distinct, T is a composite of two perspectivities $T_1 : l \rightarrow m, T_2 : m \rightarrow l'$ with centres O_1, O_2 respectively. Then $K = T(K) = T_2(T_1(K)) = (O_2 \vee T_1(K)) \wedge m$, so $K \in O_2 \vee T_1(K)$, so $K, O_2, T_1(K)$ are collinear. But $T_1(K) = (O_1 \vee K) \wedge m$ so $K, O_1, T_1(K)$ are collinear. So O_1, O_2, K are collinear. Let $L = l \wedge m, L' = l' \wedge m, O = (O_1 \vee L') \wedge (O_2 \vee L)$. Let $S : l \rightarrow l'$ be the perspectivity from l to l' with centre O . We will show that $T = S$, a perspectivity.



Take any P on ℓ . Then we have the array

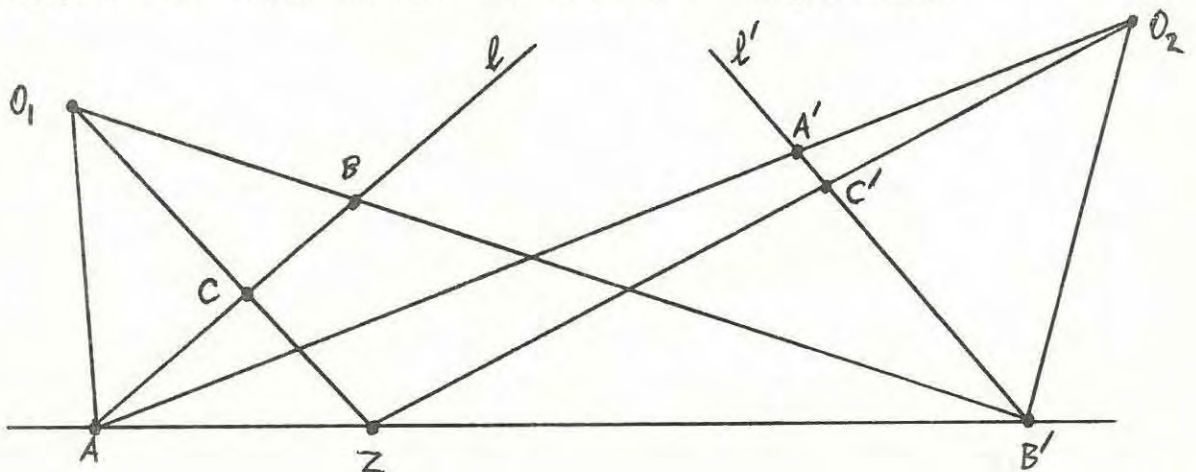
$$\left\| \begin{array}{ccc} O_1 & K & O_2 \\ L & T_1(P) & L' \\ P & O & T(P) \end{array} \right\|$$

so $P, O, T(P)$ are collinear, so $S(P) = T(P)$. //

The Fundamental Theorem: In a Pappusian plane, there is exactly one projectivity which takes three given distinct points of one line to three given distinct points of another (perhaps the same) line.

Proof: Suppose A, B, C are distinct points on a line ℓ , and A', B', C' are distinct points on a line ℓ' .

(a) Consider the case where $\ell \neq \ell'$. Let m be the line $A \vee B'$ (provided A, B' are distinct). Let Z be any point on m other than A, B' . Let $O_1 = (B \vee B') \wedge (C \vee Z)$, $O_2 = (A' \vee A) \wedge (C' \vee Z)$, and let $T_1: \ell \rightarrow m$, $T_2: m \rightarrow \ell'$ denote the perspectivities with centres O_1, O_2 respectively. Then $T_2T_1: \ell \rightarrow \ell'$ takes A, B, C to A', B', C' respectively.



So we have at least one projectivity with the desired property (a result already proven). Suppose $T: \ell \rightarrow \ell'$ is any projectivity which takes A, B, C to A', B', C' . Then $T_2^{-1}T: \ell \rightarrow m$ takes A, B, C to A, B', Z ; $A = \ell \wedge m$. By the last theorem, $T_2^{-1}T: \ell \rightarrow m$ is a perspectivity and its centre must be $(B \vee B') \wedge (C \vee Z) = O_1$. So $T_2^{-1}T = T_1: \ell \rightarrow m$.

So $T = T_1T_2: \ell \rightarrow \ell'$.

(b) Suppose $\ell = \ell'$. Let ℓ_0 be any line distinct from ℓ and let $T_0: \ell \rightarrow \ell_0$ be any perspectivity from ℓ to ℓ_0 . Put $A_0 = T_0(A)$, $B_0 = T_0(B)$, $C_0 = T_0(C)$. By the last part of the proof, there exists a unique projectivity $T': \ell_0 \rightarrow \ell$ taking A_0, B_0, C_0 to A', B', C' . Then $T'T_0: \ell \rightarrow \ell$ takes A, B, C to A', B', C' . Suppose $T: \ell \rightarrow \ell$ also does this. Then $T T_0^{-1}: \ell_0 \rightarrow \ell$ takes A_0, B_0, C_0 to A', B', C' whereas T' was unique with this property. So $T T_0^{-1} = T'$ and $T = T'T_0$. //

Corollary: If a projectivity on a line in a Pappusian plane has three fixed points then it is the identity.

Proof: Suppose $T: \ell \rightarrow \ell$ is a projectivity which takes distinct A, B, C to A, B, C respectively. By the fundamental theorem there is a unique projectivity which does this: the identity does it. So T is the identity. //

If the points $A, B, C, D, \dots$ correspond to the points $A', B', C', D', \dots$ respectively under some projectivity then we write

$$A B C D \dots \overline{\wedge} A' B' C' D' \dots$$

By the fundamental theorem, if A, B, C are distinct collinear points and A', B', C' are distinct collinear points then

$$A B C \overline{\wedge} A' B' C'$$

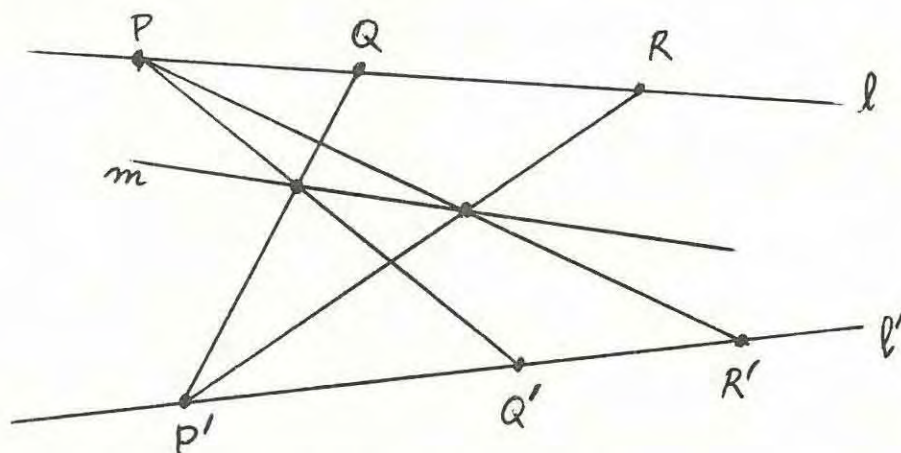
fully specifies a projectivity.

Theorem: In a Pappusian plane, if $T: \ell \rightarrow \ell'$ is a projectivity there is a line m (called the cross-axis of T) such that, for each pair of distinct points A, B on ℓ , the point $(A \vee T(B)) \wedge (T(A) \vee B)$ (if A or B is a fixed point of T this point is not well defined) is on m .

Proof: If $\ell = \ell'$ then $m = \ell$ is the meaning given to the theorem. Otherwise, let $T: \ell \rightarrow \ell'$ be the projectivity determined by

$$P Q R \overline{\wedge} P' Q' R'$$

where P, Q, R are distinct and P is not on ℓ or ℓ' . Let m be the line $((P \vee Q') \wedge (P' \vee Q)) \vee ((P \vee R') \wedge (P' \vee R))$.

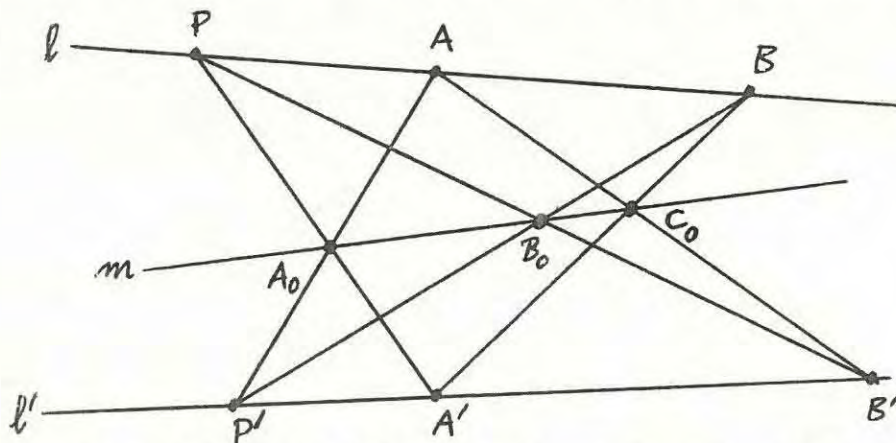


Consider the transformation $S : \ell \rightarrow \ell'$ given by:

$$S(X) = ((P' \vee X) \wedge m) \vee P \wedge \ell' \text{ for } X \text{ on } \ell.$$

Then $S(P) = P'$, $S(Q) = Q'$, $S(R) = R'$ and S is a projectivity since it is the composite of the perspectivity from ℓ to m determined by P' with the perspectivity from m to ℓ' determined by P . By the fundamental theorem then, $T = S$.

Now take any pair of distinct points A, B on ℓ (which are not fixed by S).



Let $A' = S(A)$, $B' = S(B)$, $A_0 = (P \vee A') \wedge (P' \vee A)$, $B_0 = (P \vee B') \wedge (P' \vee B)$, $C_0 = (A \vee B') \wedge (A' \vee B)$. Then we have the array

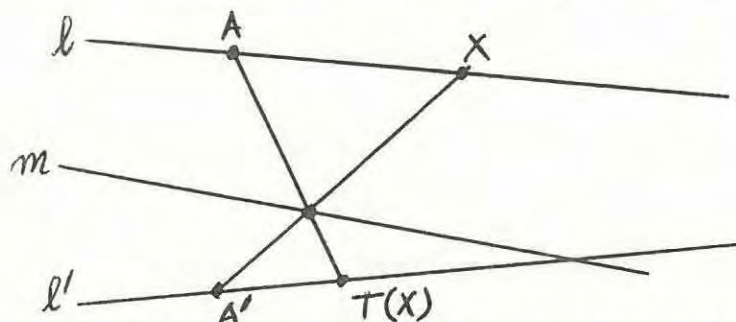
$$\begin{vmatrix} P & A & B \\ P' & A' & B' \\ C_0 & B_0 & A_0 \end{vmatrix},$$

so C_0, B_0, A_0 are collinear; that is, C_0 is on $A_0 \vee B_0 = m$. This is the required result since $C_0 = (A \vee T(B)) \wedge (T(A) \vee B)$.

This proof also shows that a projectivity $T : \ell \rightarrow \ell'$ between distinct

lines is determined by its cross-axis m and any pair of corresponding points A, A' (avoiding the points $\ell \wedge m, \ell' \wedge m, \ell \wedge \ell'$); namely,

$$T(X) = (((A' \vee X) \wedge m) \vee A) \wedge \ell'.$$

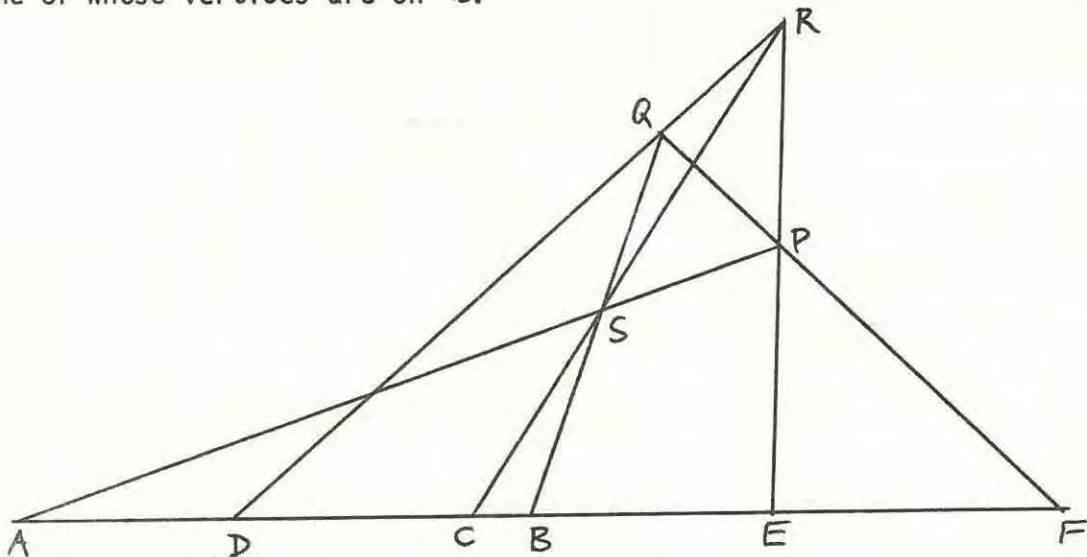


In a Pappusian plane we have proven the fundamental theorem. The fundamental theorem implies Pappus's axiom (exercise; see Meserve p.62). So in a Desarguesian plane, the fundamental theorem and Pappus's axiom are logically equivalent.

§8. Projectivities on a line

Suppose we have a Desarguesian plane and a fixed line ℓ in it.

Six points on ℓ form a *quadrangular set* if they are the points of intersection of the line ℓ with the six sides of a complete quadrangle none of whose vertices are on ℓ .



Given five points A, B, C, D, E , a sixth point F may be constructed by choosing a quadrangle $PQR S$ such that $(P \vee S) \wedge \ell = A$, $(Q \vee S) \wedge \ell = B$, $(R \vee S) \wedge \ell = C$, $(R \vee Q) \wedge \ell = D$, $(R \vee P) \wedge \ell = E$. Then $F = (P \vee Q) \wedge \ell$.

As for the harmonic construction, the point F does not depend on the choice of the quadrangle $PQR S$. The student should draw a diagram

with two choices $PQR S$ and $P'Q'R'S'$ of the quadrangle (the diagram perhaps looks neatest if one quadrangle is above ℓ and the other is below). To prove this independence we proceed as follows:

$$\Delta PQR S \stackrel{\vee}{\underset{CAE}{=}} \Delta P'Q'R'S' \implies P \vee P', R \vee R', S \vee S' \text{ concurrent.}$$

$$\Delta QRS \stackrel{\vee}{\underset{CBD}{=}} \Delta Q'Q'R'S' \implies Q \vee Q', R \vee R', S \vee S' \text{ concurrent.}$$

Let O denote the point common to $P \vee P', Q \vee Q', R \vee R', S \vee S'$ ($PQR S$ and $P'Q'R'S'$ are "perspective quadrangles").

$$\Delta PQR \stackrel{O}{\underset{\wedge}{=}} \Delta P'Q'R' \implies D, E, (P \vee Q) \wedge (P' \vee Q') \text{ collinear.}$$

So $P \vee Q, P' \vee Q'$ meet on $D \vee E = \ell$; namely, at the point F .

Thus F is completely determined by A, B, C, D, E . Under these circumstances we write

$$ABC \underline{Q} DEF.$$

This is clearly unchanged if we apply any permutation to ABC and the same permutation to DEF . It is also equivalent to each of

$$AEF \underline{Q} DBC, DBF \underline{Q} AEC, DEC \underline{Q} ABF.$$

Theorem: In a Pappusian plane, $ABC \underline{Q} DEF$ implies $DEF \underline{Q} ABC$.

Proof: With P, Q, R, S as above, let

$$R' = (Q \vee R) \wedge (S \vee F), \quad P' = (P \vee S) \wedge (Q \vee C)$$

and apply Pappus's axiom to the array

$$\left\| \begin{array}{ccc} P & F & Q \\ C & R & S \\ R' & P' & E \end{array} \right\| ,$$

(draw the diagram!) so that we see that $R' \vee P'$ passes through E . So the quadrangle $P'Q'R'S$ yields $ABF \underline{Q} DEC$. This is equivalent to $DEF \underline{Q} ABC$ //

Harmonic conjugation is a special case:

Theorem: $AB \underline{H} CF$ is equivalent to $ABC \underline{Q} ABF$ //

We restrict attention to a Pappusian plane for the remainder of this section.

As a corollary of the fundamental theorem we saw that a projectivity from ℓ to ℓ which leaves three points fixed leaves all the points fixed (that is, it is the identity transformation). A projectivity from a line to itself is said to be *elliptic*, *parabolic*, or *hyperbolic* according as it has no, one, or two fixed points respectively.

A projectivity $A B F \xrightarrow{\quad} A C E$ with a given fixed point A may be constructed as follows. Let S, P be any two points collinear with A . Put $Q = (B \vee S) \wedge (F \vee P)$, $R = (C \vee S) \wedge (E \vee P)$, $D = (Q \vee R) \wedge \ell$. Then we have perspectivities

$$A B F \xrightarrow{Q} A S P \xrightarrow{R} A C E;$$

so the composite is a projectivity with A as a fixed point. The point D is also fixed. So this construction gives a hyperbolic projectivity when $A \neq D$ and a parabolic projectivity when $A = D$. Note that the last diagram applies, so $A B C \xrightarrow{Q} D E F$. (Since there is only one projectivity determined by $A B F \xrightarrow{\quad} A C E$, this gives another proof - in a Pappusian plane - that the construction of D is independent of the choice of S and P).

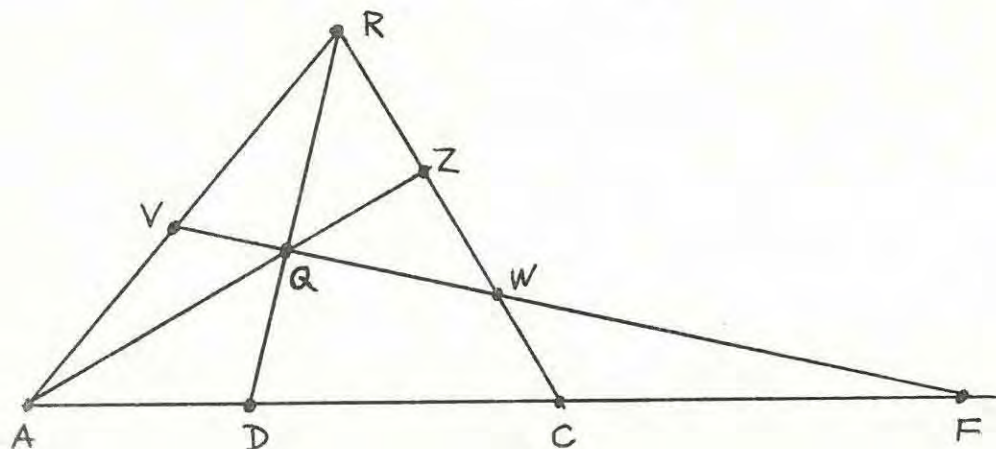
A transformation from a set to itself is called an *involution* when the transformation is its own inverse. A transformation $T: \ell \rightarrow \ell$ is an involution means that $T(X) = Y$ if and only if $T(Y) = X$. Two points $X Y$ are called a *pair of the involution* when $T(X) = Y$; so if $X Y$ is a pair then $Y X$ is a pair; order in the pair is irrelevant. A *projective involution* is a projectivity which is also an involution.

Theorem: Any projectivity on a line ℓ which interchanges two distinct points is an involution.

Proof: Suppose A, D are distinct points and $T: \ell \rightarrow \ell$ is a projectivity such that $T(A) = D$ and $T(D) = A$. We want to prove that T is an involution. Take C on ℓ and $T(C) = F$. (We must prove that $T(F) = C$). Construct R not on ℓ , and any line m through F distinct from ℓ . Let $V = (R \vee A) \wedge m$, $Q = (R \vee D) \wedge m$, $W = (R \vee C) \wedge m$, $Z = (A \vee Q) \wedge (R \vee C)$.

Then

$$A D C F \xrightarrow{Q} Z R C W \xrightarrow{A} Q V F W \xrightarrow{R} D A F C$$



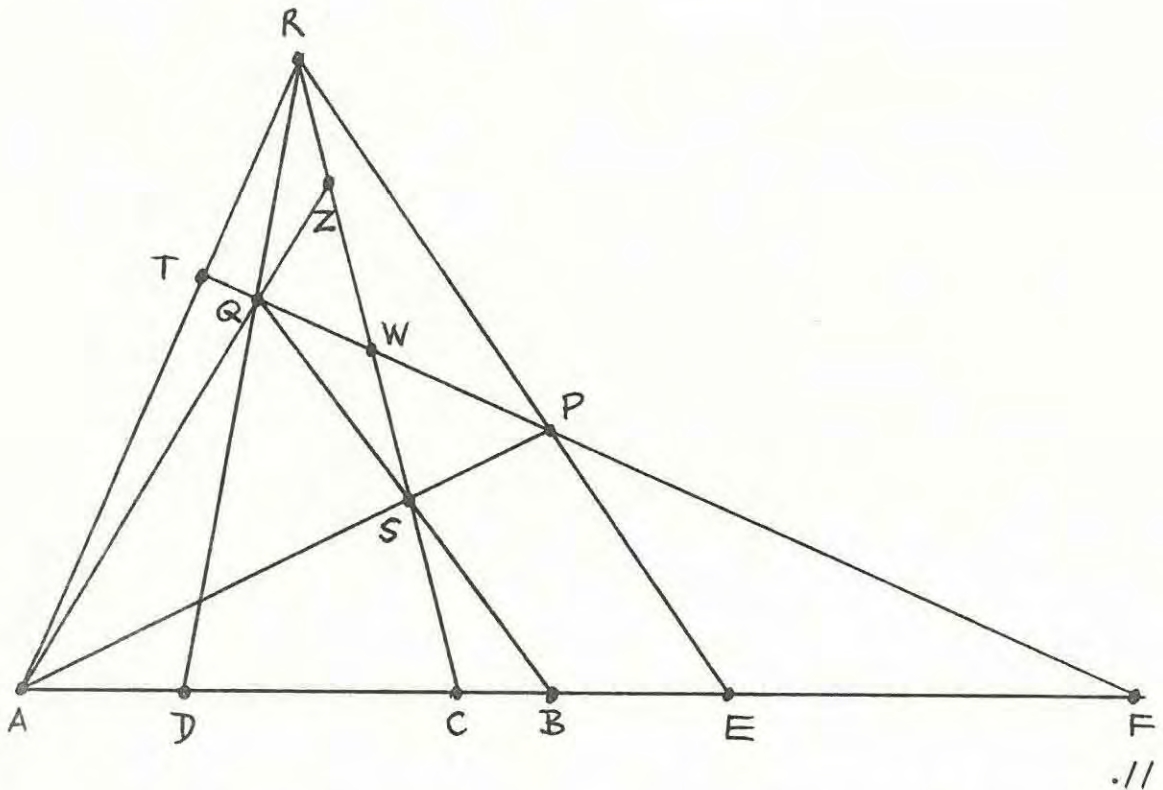
This composite of three perspectivities must be T by the fundamental theorem since $T(A) = D$, $T(D) = A$ and $T(C) = F$. So $T(F) = C$. //

Theorem: The three pairs of opposite sides of a quadrangle meet any line (not through a vertex) in three pairs of a projective involution.

Proof: Suppose we have $A B C Q D E F$ using the quadrangle $P Q R S$. Construct $Y = (Q \vee F) \wedge (R \vee A)$, $Z = (A \vee Q) \wedge (R \vee C)$, $W = (V \vee F) \wedge (R \vee C)$. Y, V are both supposed to be T .

Then $A D C F B \stackrel{Q}{\wedge} Z R C W S \stackrel{A}{\wedge} Q T F W P \stackrel{R}{\wedge} D A F C E$.

This composite of three perspectivities is an involution since it interchanges A and D (and also C and F). So $A D$, $C F$, $B E$ are pairs of an involution.



A projective involution $A C D \bar{\wedge} D F A$ is determined by two of its pairs $A D$, $C F$ (or any other two of its pairs), so we denote the involution by $(AD)(CF)$ (or $(DA)(CF)$, or $(CF)(AD)$ etc.).

Theorem: If a projective involution has one fixed point then it has another, and the pairs of the involution are the pairs of harmonic conjugates with respect to these two fixed points.

Proof: Suppose we have a projective involution $(AA)(CF)$ with a fixed point A . Let B be the point such that $A B \bar{\wedge} C F$. Then $A B C Q A B F$, so

BB is a pair of the involution. So B is another fixed point. The involution may also be written as (AA)(BB). Harmonic conjugation with respect to A and B does give a projective involution with A, B as fixed points since $ABH C' F'$ is equivalent to $ABC' Q ABF'$. So (AA)(BB) is the involution obtained by harmonic conjugation with respect to A and B.//

Corollary: *There are no parabolic projective involutions.*//

Hyperbolic projective involutions are just harmonic conjugation with respect to their fixed points. Parabolic projective involutions do not exist. Elliptic projective involutions will be important for producing Euclidean similarity geometry. In some projective planes (even Pappusian planes) elliptic projective involutions do not exist. Algebraically their existence depends on the existence of quadratics which do not have roots.

§9. Conics

Euclid considered other figures besides points and lines - circles for example. A circle for Euclid was the set of points equidistant from some fixed point. In a projective plane there is no concept of distance - only points, lines and incidence between them. It should not be expected that circles should be describable in terms of points, lines and incidence since these things are preserved if we "slant the page", whereas a circle becomes an "ellipse". Moreover, the Euclidean plane is not quite a projective plane (as we saw in §3 an extra line had to be added). So it is feasible that, to obtain something like the Euclidean plane from a projective plane, a line should be taken out; and then the line should be thought of as being "at infinity". If we take out a line which meets the "ellipse" in one place and take the line "off to infinity", the resulting figure would look like a "parabola". If we take out a line which meets the "ellipse" in two places, and take the line "off to infinity", the resulting figure would look like a "hyperbola". Those figures which are either ellipses, parabolae or hyperbolae are called *conics*.

This introduction is not meant to be formal, only to provide a way of thinking: yet it will be made precise. We hope that it provides also an intuitive feeling for why we cannot distinguish the various types of conics in a projective plane - why all conics look alike in a projective plane. The amazing result, however, is that *conics can be defined* in a projective plane, and many of their well-known properties can be deduced in a Pappusian plane.

In this section then, we work in a Pappusian plane.

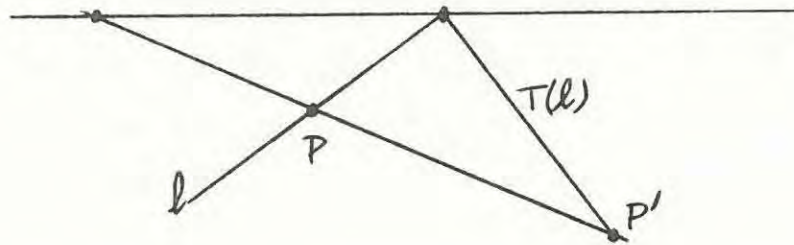
The remarkable observation of Steiner (1832) is that a projectivity between pencils of lines has a conic associated with it, and, in fact, conics may be defined in terms of such projectivities.

A *conic* is the set of points obtained as the intersection of corresponding lines of a projectivity (which is not a perspectivity) between

the pencils of lines through two (distinct) points. If either the projectivity is a perspectivity or the two points are not distinct then the intersections of corresponding lines is called a *degenerate* (or *reducible*) conic.

In symbols, if $T : P \rightarrow P'$ is a projectivity from the pencil of lines through P to the pencil of lines through P' then the conic ω_T associated with T is the set of points $\ell \wedge T(\ell)$ where ℓ is a line through P . The conic is degenerate or not according as P, P' are equal or not, or T is a perspectivity or not.

Let us examine the degenerate cases. Suppose $P = P'$. Then we are in the dual situation to §8. If $T : P \rightarrow P$ is elliptic (there are no lines ℓ through P such that $T(\ell) = \ell$), then for each line ℓ through P , $\ell \wedge T(\ell) = P$; so P is the only point on the degenerate conic. If $T : P \rightarrow P$ does have a fixed line ℓ (that is, $T(\ell) = \ell$) then $\ell \wedge T(\ell) = \ell \wedge \ell$ is the set of points on ℓ ; so all the points on ℓ are on the conic. Hence, if T is a parabolic projectivity the conic consists of the points of one line, and if T is a hyperbolic projectivity the conic consists of the points of two lines. If $T : P \rightarrow P$ is the identity projectivity then every line through P is fixed and every point of the plane lies on some line through P , so every point of the plane is on the conic. Suppose $P \neq P'$ and $T : P \rightarrow P'$ is a perspectivity with axis o ; so $T(\ell) = (\ell \wedge o) \vee P'$.



Then, for any line ℓ through P other than $P \vee P'$, $\ell \wedge T(\ell)$ is on o , and every point on o other than $(P \vee P') \wedge o$ is of this form. If $\ell = P \vee P'$ then $T(\ell) = \ell$. Hence the conic consists of all the points on o and all the points on $P \vee P'$. So degenerate conics fall into four types:

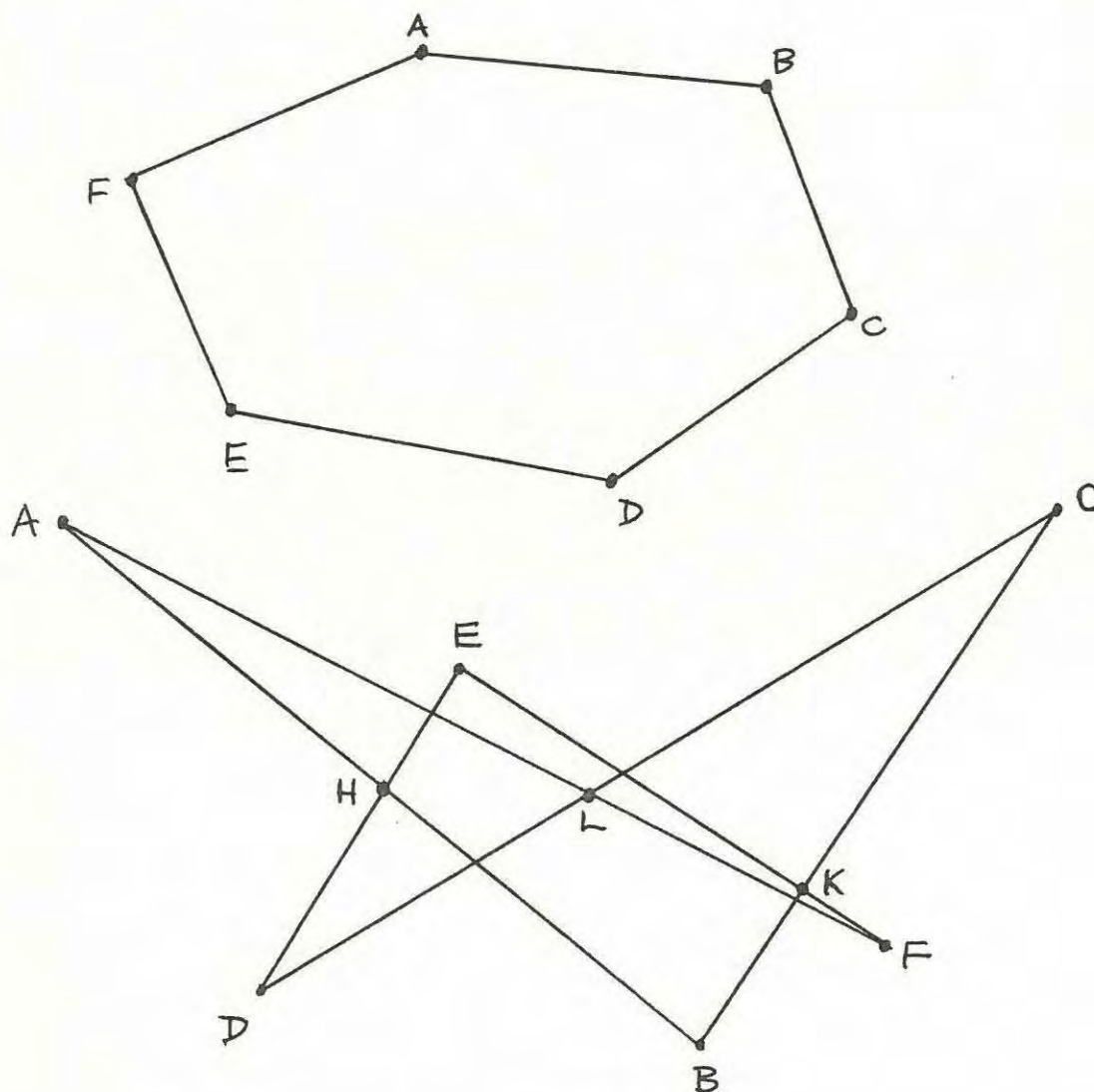
1. *trivial conic* : consists of just one point;
2. *line pair* : consists of the points on two distinct lines;
3. *repeated line* : consists of the points on one line;
4. *illusory conic* : consists of all the points of the plane.

Now we shall discuss non-degenerate conics in some detail.

Essential to the discussion is the concept of a hexagon. A *hexagon* is an ordered set of six distinct points, called *vertices*, where the same hexagon is given by a cyclic permutation of the vertices. There are 60 hexagons

which can be formed from six given points. The *sides* are the lines obtained by cyclic joining of the vertices. The sides of a hexagon $A B C D E F$ (which is the same as the hexagon $B C D E F A$, but not the same as $B A C D E F$) come in *opposite pairs* as follows:

$A \vee B$ is opposite $D \vee E$,
 $B \vee C$ is opposite $E \vee F$,
 $C \vee D$ is opposite $F \vee A$.



The intersections of opposite sides give the three *diagonal points* H, K, L of the hexagon.

Pappus's axiom can be expressed in the tidy form:

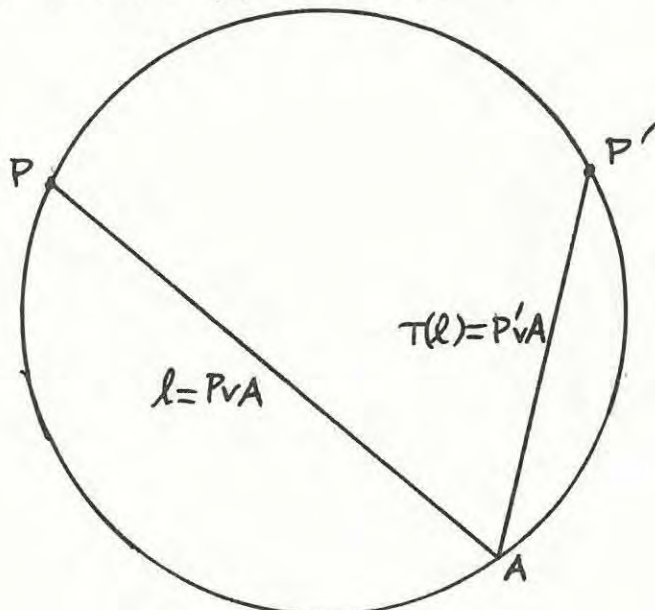
if alternate vertices of a hexagon lie on two lines, then the diagonal points are collinear.

This axiom may be thought of as a degenerate case of a theorem we shall prove due to Pascal.

The next lemma shows that the projectivity $T : P \rightarrow P'$ can be recaptured from the conic ω_T and the two points P, P' .

Lemma: If $T : P \rightarrow P'$ is a projectivity then a point A distinct from P, P' is on ω_T exactly when $T(P \vee A) = P' \vee A$.

Proof: Suppose $A \neq P, P'$. Then A is on ω_T exactly when $A = \ell \wedge T(\ell)$ for some ℓ through P . Then $\ell = P \vee A$ and $T(\ell) = P' \vee A$. So A is on ω_T exactly when $P' \vee A = T(P \vee A)$.



.//

Lemma: If $T : P \rightarrow P'$ is a projectivity then $\omega_T = \omega_{T^{-1}}$

Proof: If ℓ goes through P , let $\ell' = T(\ell)$, then $\ell \wedge T(\ell) = \ell' \wedge T^{-1}(\ell')$. So the points on ω_T are exactly the points on $\omega_{T^{-1}}$. //

A line which meets a non-degenerate conic in exactly one point is called a *tangent*. Tangents for degenerate conics are as follows:

the tangents to a trivial conic consisting of one point P are the lines through P ;

the tangents to a line pair ℓ, m are all lines through the point $\ell \wedge m$;

all lines in the plane are tangents to a repeated line or an illusory conic.

A point of contact of a tangent to a conic is a point on both the tangent and the conic.

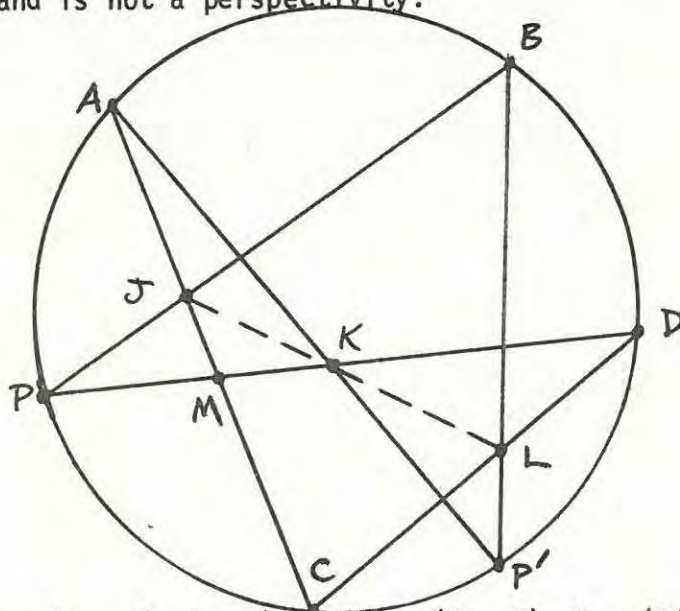
Lemma: If ω_T is a non-degenerate conic arising from the projectivity $T: P \rightarrow P'$ then P, P' are on ω_T and $T^{-1}(P \vee P')$, $T(P \vee P')$ are tangents to ω_T with points of contact P, P' respectively.

Proof: Let $p = P \vee P'$. Since T is not a perspectivity, $T(p) \neq p$. But $T(p), p$ both go through P' , so $P' = p \wedge T(p)$ is on ω_T . Suppose $T(p)$ meets ω_T at $Q \neq P'$. Since Q is on ω_T , $T(P \vee Q) = P' \vee Q$. Since Q is on $T(p)$, $T(p) = P' \vee Q$. So $T(p) = T(P \vee Q)$. So $P \vee Q = p$. So Q is on p ; it is also on $T(p)$, so $Q = p = T(p) = P'$ - a contradiction. Hence $T(p)$ is a tangent to ω_T with P' as its point of contact.

Applying this result to $T^{-1}: P' \rightarrow P$ and using $\omega_{T^{-1}} = \omega_T$, we obtain the result that $T^{-1}(p)$ is a tangent to ω_T with P as its point of contact. //

Lemma: Suppose ω is a conic arising from a projectivity from P to P' . If A, B, C, D are on ω then the diagonal points of the hexagon $P B P' A C D$ are collinear.

Proof: The degenerate cases are either trivial or covered by Pappus's axiom. We thus may suppose $\omega = \omega_T$ where $T: P \rightarrow P'$ is a projectivity between distinct points and is not a perspectivity.



Let $J = (P \vee B) \wedge (A \vee C)$, $L = (B \vee P') \wedge (C \vee D)$, $K = (P' \vee A) \wedge (D \vee P)$, $M = (A \vee C) \wedge (P \vee D)$, $N = (A \vee P') \wedge (D \vee C)$. The projectivity $T: P \rightarrow P'$ may be written

$$P \vee (A B C D) \underline{\vee} P' \vee (A B C D),$$

which is the same thing as

$$P \vee (A J C M) \underline{\vee} P' \vee (N L C D)$$

since $P \vee J = P \vee B$, $P \vee D = P \vee M$, $P' \vee A = P' \vee N$, $P' \vee B = P' \vee L$. Moreover A, J, C, M and N, L, C, D are two sets of four collinear points. So, by the corollary to a theorem of §4,

$$A J C M \bar{\vee} N L C D.$$

For this projectivity C is a fixed point, so by a theorem of §7, the projectivity is a perspectivity. So $A \vee N$, $J \vee L$, $M \vee D$ are concurrent; that is, $J \vee L$ passes through $K = (A \vee N) \wedge (M \vee D)$. So J, K, L are collinear. //

Lemma: If the diagonal points of the hexagon $P B P' A C D$ are collinear then A, B, C, D are on a conic determined by a projectivity from P to P' .

Proof: Let J, K, L be the diagonal points of the hexagon as before, and take M, N as before. Then

$$\begin{aligned} P \vee (A B C D) &\bar{\vee} K \vee (A J C M) \\ &A J C M \\ &\vee \\ &= P' \vee (A B C D). \\ &N L C D \end{aligned}$$

If $T : P \rightarrow P'$ is the composite of these two perspectivities, then ω_T is the required conic. //

Theorem: A (non-degenerate) conic is uniquely determined by five distinct points (no three of which are collinear) on it.

Proof: The proof will be broken up into steps.

Step 1: Given five distinct points, P, P', A, B, C there is exactly one conic $\omega(P P'; A B C)$ which is determined by a projectivity from P to P' and goes through A, B, C .

For: If A, B, C are on a conic determined by a projectivity from P to P' then

$$P \vee (A B C) \bar{\vee} P' \vee (A B C),$$

and, by the fundamental theorem (dual), this completely determines the projectivity, and hence the conic.

Step 2: If the point D is on $\omega(P P'; A B C)$ then $\omega(P P'; A B D) = \omega(P P'; A B C)$.

For: If D is on $\omega(P P'; A B C)$ then

$$P \vee (A B C D) \bar{\vee} P' \vee (A B C D)$$

and this projectivity is completely determined by

$$P \vee (A B D) \bar{\vee} P' \vee (A B D)$$

which is the projectivity determining $\omega(P P'; A B D)$.

Step 3: $\omega(P P'; A B C) = \omega(B A; C P' P)$.

For: Take any point D on $\omega(P P'; A B C)$. By the second last lemma, the diagonal points of the hexagon $P B P' A C D$ are collinear. These are the same as the diagonal points of $B P' A C D P$. By the last lemma then, C, P', D, P are on a conic determined by a projectivity from B to A . By Step 1 this conic must be $\omega(B A; C P' P)$. So D is on $\omega(B A; C P' P)$. This proves that every point of $\omega(P P'; A B C)$ is on $\omega(B A; C P' P)$. Similarly every point of $\omega(B A; C P' P)$ is on $\omega(P P'; A B C)$, hence the result.

Step 4: We now prove the theorem. The degenerate cases are easily dismissed, so suppose P, P', A, B, C are five points no three collinear. One conic through all five is $\omega(P P'; A B C)$ as in Step 1. Suppose ω is any conic through P, P', A, B, C . Then ω is determined by a projectivity from Q to Q' say. By Step 1, since A, B, C are on ω , ω must be $\omega(Q Q'; A B C)$. So

$$\begin{aligned}\omega &= \omega(Q Q'; A B C) \\ &= \omega(B A; C Q' Q) \text{ by Step 3,} \\ &= \omega(B A; C Q' P) \text{ by Step 2 since } P \text{ is on } \omega, \\ &= \omega(B A; C P Q') \text{ by Step 1,} \\ &= \omega(B A; C P P') \text{ by Step 2 since } P' \text{ is on } \omega, \\ &= \omega(P P'; A B C) \text{ by Step 3.}\end{aligned}$$

So $\omega(P P'; A B C)$ is the only conic through P, P', A, B, C . //

Pascal's Mystic Hexagram Theorem (1640): *If the vertices of a hexagon are points of a conic, its diagonal points are collinear.*

Proof: Suppose the vertices of the hexagon $P P' A B C D$ are on the conic ω . The five points P, P', A, B, C determine a projectivity $T: P \rightarrow P'$ given by

$$P \vee (A B C) \quad \underline{\vee} \quad P' \vee (A B C),$$

and ω_T goes through P, P', A, B, C . By the last theorem then $\omega_T = \omega$. By an earlier lemma, since A, B, C, D are on ω_T the diagonal points of the hexagon $P B P' A C D$ are collinear. //

What we have called a conic might well be called a *point conic*. The dual of a point conic is a *line conic*. So a line conic is the set of lines obtained as the joins of corresponding points of a projectivity between the pencils of points on two lines. A *point of contact* of a line conic is a point which is on exactly one line of the line conic.

It can be proven geometrically (we shall not do this here) that the set of tangents to a point conic is a line conic; and dually, the set of points of contact of a line conic is a point conic.

Just as we defined a point triangle and a line triangle and we saw how one completely determined the other and led to thinking of a triangle as the point triangle with its associated line triangle, we may think of a conic as the point conic together with its associated line conic.

CHAPTER II : ALGEBRAIC PROJECTIVE GEOMETRY

§1. Coordinates for a Desarguesian plane.

Geometric problems of the Euclidean plane become algebraic when coordinates are introduced into the plane. Each point in the plane is represented by a pair (x, y) of real numbers x, y . We shall outline the rigorous way in which coordinates can be introduced into a Desarguesian plane; however, the coordinates need not be real numbers, just elements of some skew field. Skew fields do have many of the properties of the real numbers.

A *skew field* is a set F together with two binary operations "+" and "x" satisfying the following axioms.

Closure: for x, y in F , $x + y$ and $x \times y$ are in F ($x \times y$ is written xy);

Associativity: for x, y, z in F ,

$$(x + y) + z = x + (y + z) \text{ and } (xy)z = x(yz).$$

Identities: there are elements $0, 1$ in F such that $x + 0 = 0 + x = x$ and $x1 = 1x = x$ for all x in F ;

Inverses: for each x in F there is an element $-x$ in F such that $x + (-x) = (-x) + x = 0$, and, for each x in F other than 0 there is an element $x^{-1} = \frac{1}{x}$ in F such that $xx^{-1} = x^{-1}x = 1$;

Distributivity: for x, y, z in F ,

$$x(y + z) = xy + xz \text{ and } (x + y)z = xz + yz.$$

Commutativity of +: for x, y in F ,

$$x + y = y + x.$$

If the further axiom:

Commutativity of x: for x, y in F ,

$$xy = yx ;$$

is satisfied then F is called a *field*.

Notice in particular that the complex numbers $\mathbb{C}$, the real numbers $\mathbb{R}$, the rational numbers $\mathbb{Q}$, and, for each prime number p , the integers modulo p (denoted by $\mathbb{Z}_p$) are all examples of fields. The quaternions $\mathbb{H}$ (after Hamilton) give an example of a skew field which is not a field.

The real numbers do have properties which cannot be deduced from the field axioms, namely, the order property and the completeness property (in the presence of the order, completeness amounts to the fact that each non-empty set of real numbers which is bounded above has a least upper bound).

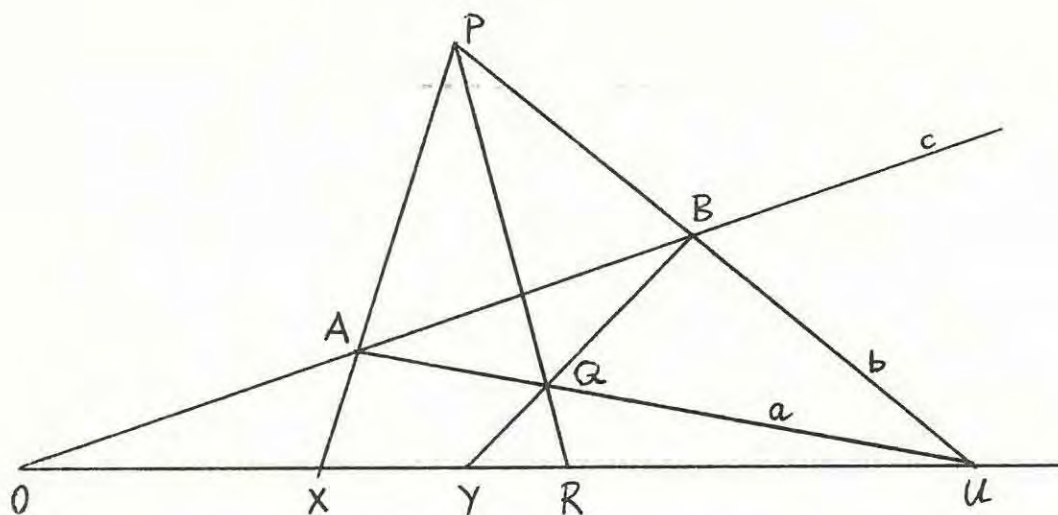
Suppose now we have a Desarguesian plane. We shall show that if one point is taken away from a line in the plane then the remaining points on the line form a skew field. (This is just what happens in the case of the Euclidean plane plus a line since the points on any line, leaving out the ideal point, are in 1-1 correspondence with the real numbers).

We have seen that all lines in the plane "have the same number of points"; that is, the points of any two lines are in 1-1 correspondence. Let ℓ be any line in the Desarguesian plane, and fix three points O, E, U on ℓ . Let F denote any set whose elements are in 1-1 correspondence with the points on ℓ other than U . Let $F^\#$ denote the set consisting of the elements of F together with the element ∞ . Then there is a 1-1 correspondence between the points of ℓ and the elements of $F^\#$, and under this correspondence, suppose the points O, E, U of ℓ correspond to $0, 1, \infty$ of $F^\#$.

The aim now is to show how F can be made into a field.

Addition: Take x, y in F . Let X, Y be the corresponding points on ℓ . Let a, b be lines through U distinct and distinct from ℓ , and let c be a line through O distinct from ℓ and meeting a, b in A, B respectively. Let

$$P = (X \vee A) \wedge b, \quad Q = (Y \vee B) \wedge a, \quad R = (P \vee Q) \wedge \ell.$$



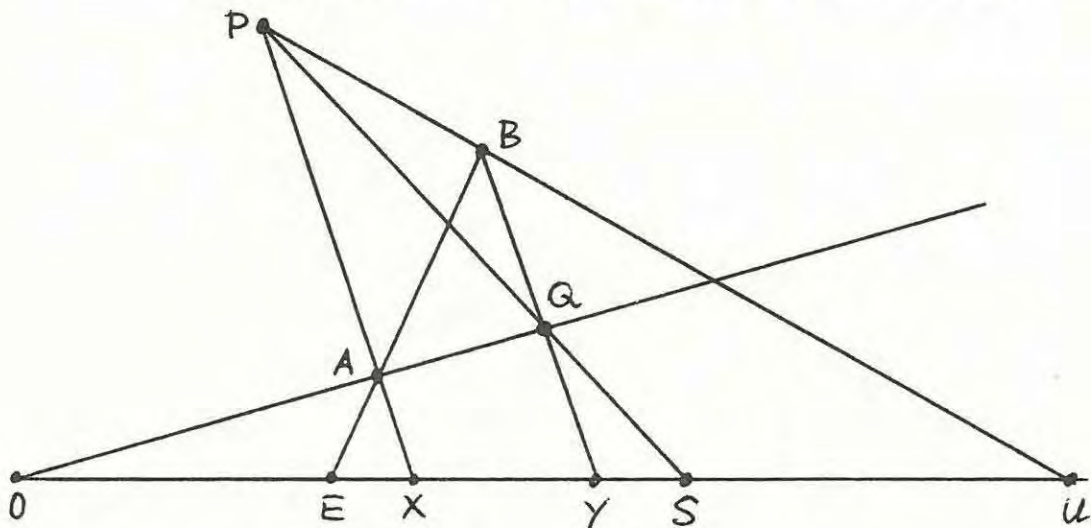
Then $UXO \underline{Q} UYR$. So R is independent of the choice of the lines a, b, c .

Thus we make the definition:

$x + y$ is the element of F corresponding to R .

Multiplication: Take x, y in F and let X, Y be the corresponding points on ℓ . Let a, b be lines through O, U respectively, and let c be a line through E meeting a in A and b in B . Let

$$P = (X \vee A) \wedge b, \quad Q = (Y \vee B) \wedge a, \quad S = (P \vee Q) \wedge \ell.$$



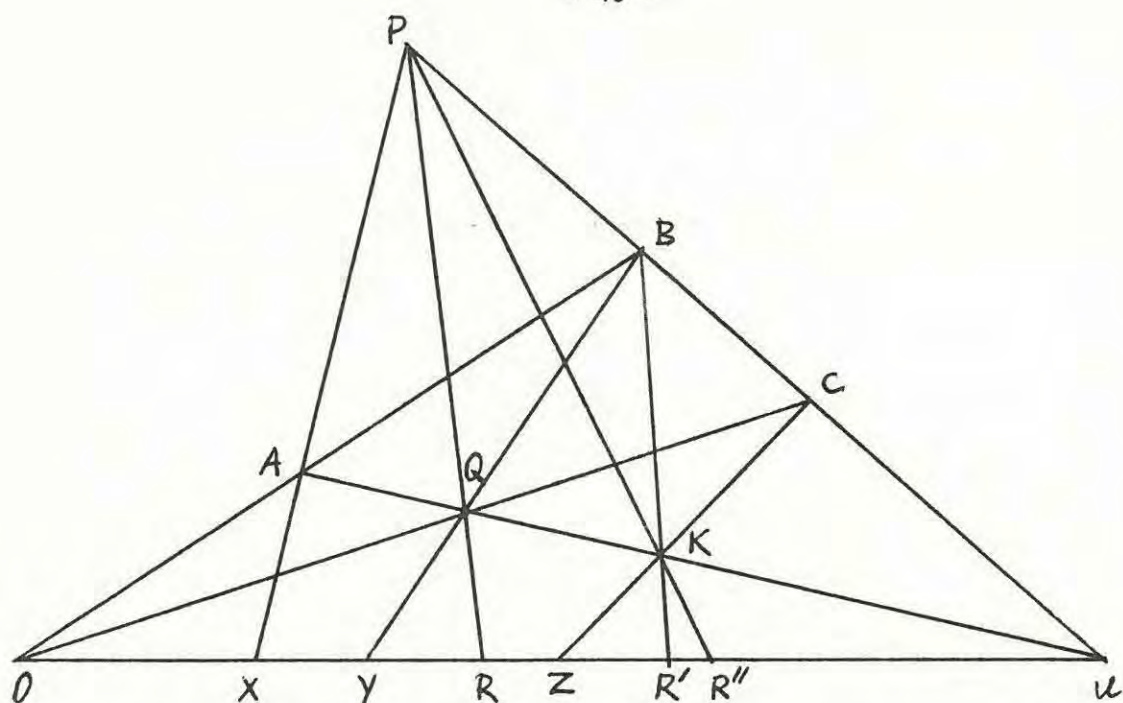
Then $O X E Q U Y S$. So S is independent of the choice of the lines a, b, c .

Thus we make the definition:

$x y$ is the element of F corresponding to S .

Theorem: Under the addition and multiplication defined above, F is a skew field.

Proof: *Associativity of addition.* With notation as in the definition of addition, let Z be another point of ℓ (corresponding to z in F). Let $(O \vee Q) \wedge b = C$, $(Z \vee C) \wedge a = K$, $(B \vee K) \wedge \ell = R'$, $(P \vee K) \wedge \ell = R''$. Using the quadrangle $B Q S C$, $U Y O Q U Z R'$, so R' corresponds to $y + z$. Using the quadrangle $P Q S C$, $U R O Q U Z R''$, so R'' corresponds to $(x + y) + z$; while using the quadrangle $P A S B$, $U X O Q U R' R'$, so R'' corresponds to $x + (y + z)$. Hence $(x + y) + z = x + (y + z)$.



The student should check that $0, 1$ corresponding to O, E are the identities of addition and multiplication and give constructions for the inverses. We leave it to the diligent student to prove associativity of multiplication and distributivity of multiplication over addition. //

Theorem: *If the plane is Pappusian then F is a field.*

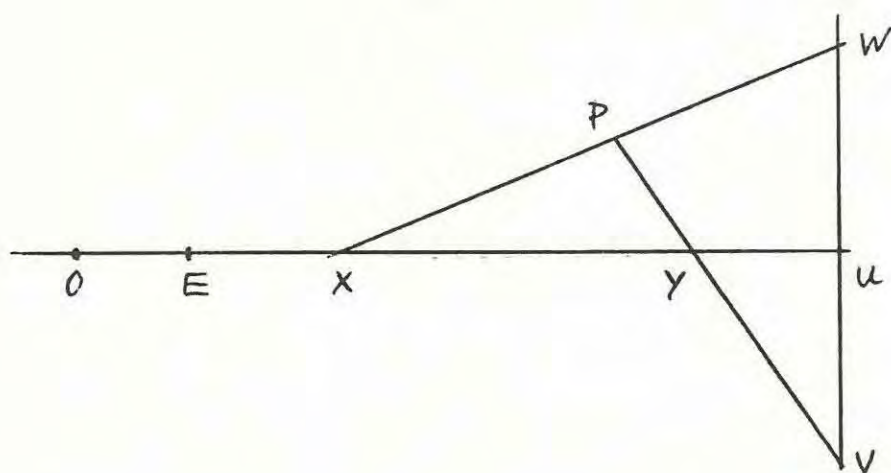
Proof: To prove commutativity of multiplication recall that in a Pappusian plane $ABC \underline{Q} DEF$ implies $DEF \underline{Q} ABC$, and this is equivalent to $AEC \underline{Q} DBF$. So $OXE \underline{Q} UYS$ implies $OYE \underline{Q} UXS$. This means S corresponds to both xy and yx . So $xy = yx$. //

In a Pappusian plane we saw that $ABC \underline{Q} DEF$ implies $(AD), (BE), (CF)$ are pairs in an involution. In fact, $ABC \underline{Q} DEF$ is equivalent to the existence of an involution of which these are pairs. So $ABC \underline{Q} DEF$ is equivalent to $ABCD \overline{E} F \overline{E} F \overline{A} B C$. If there is a projectivity $ABCD \overline{E} F \overline{A} B C$, then

$$A' B' C' D' E' F' \overline{A} B C D E F \overline{D E F A B C} \overline{D' E' F' A' B' C'},$$

so $A' B' C' \underline{Q} D' E' F'$. This proves that quadrangular sets are preserved under projectivities (we saw this for harmonic conjugates in a Desarguesian plane). So if we choose another line ℓ' and three points O', E', U' on it, then there is a unique projectivity $O E U \overline{O' E' U'}$ from ℓ to ℓ' , and since addition and multiplication are in terms of quadrangular sets, this projectivity preserves addition and multiplication constructions. Hence we arrive at essentially the same field F .

Return now to a Desarguesian plane with a line ℓ and points O, E, U on it chosen. Choose two more points V, W not on ℓ but collinear with U .



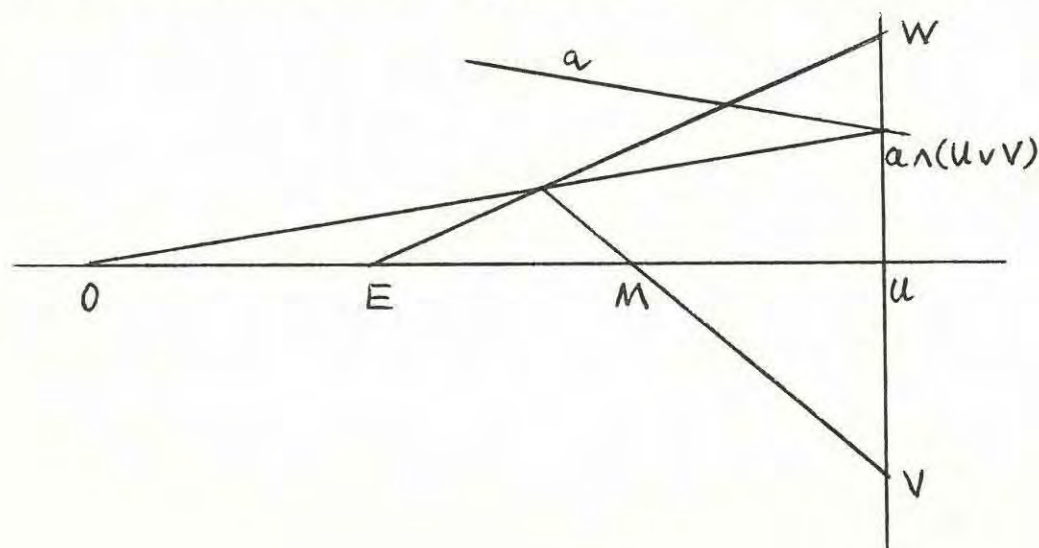
Affine coordinates: To each point P not on ℓ let X, Y denote the points where $P \vee W$ and $P \vee V$ meet ℓ . The elements x, y of F corresponding to X, Y on ℓ are called *the affine coordinates of P with respect to $\Delta O V W$ as triangle of reference and E as unit point.*

Observe that the points on $O \vee W$ have 0 as their first coordinate, the points on $O \vee V$ have 0 as their second coordinate, and the points on ℓ have their coordinates equal. We call $O \vee W$ the y -axis, $O \vee V$ the x -axis and denote $V \vee W$ by ℓ_∞ .

As remarked before, the points on any two lines are in 1-1 correspondence. Consider the particular correspondence between the points of the line $U \vee V$ and the points of the line ℓ given by the projectivity

$$U \vee W \stackrel{O}{\wedge} E \vee W \stackrel{V}{\wedge} E \vee U$$

where $I = (O \vee V) \wedge (E \vee W)$. Using this projectivity we can associate an element of $F^\#$ to each point of $U \vee V$. The *gradient* of a line a is the element of $F^\#$ associated with the point $a \wedge (U \vee V)$ in this way.



Proposition: Suppose a is a line with gradient $\mu \neq \infty$ and which meets the y -axis at the point with coordinates $(0, \gamma)$. Then those points P with coordinates (x, y) which are on a , but not on ℓ_∞ , are exactly the points satisfying the equation

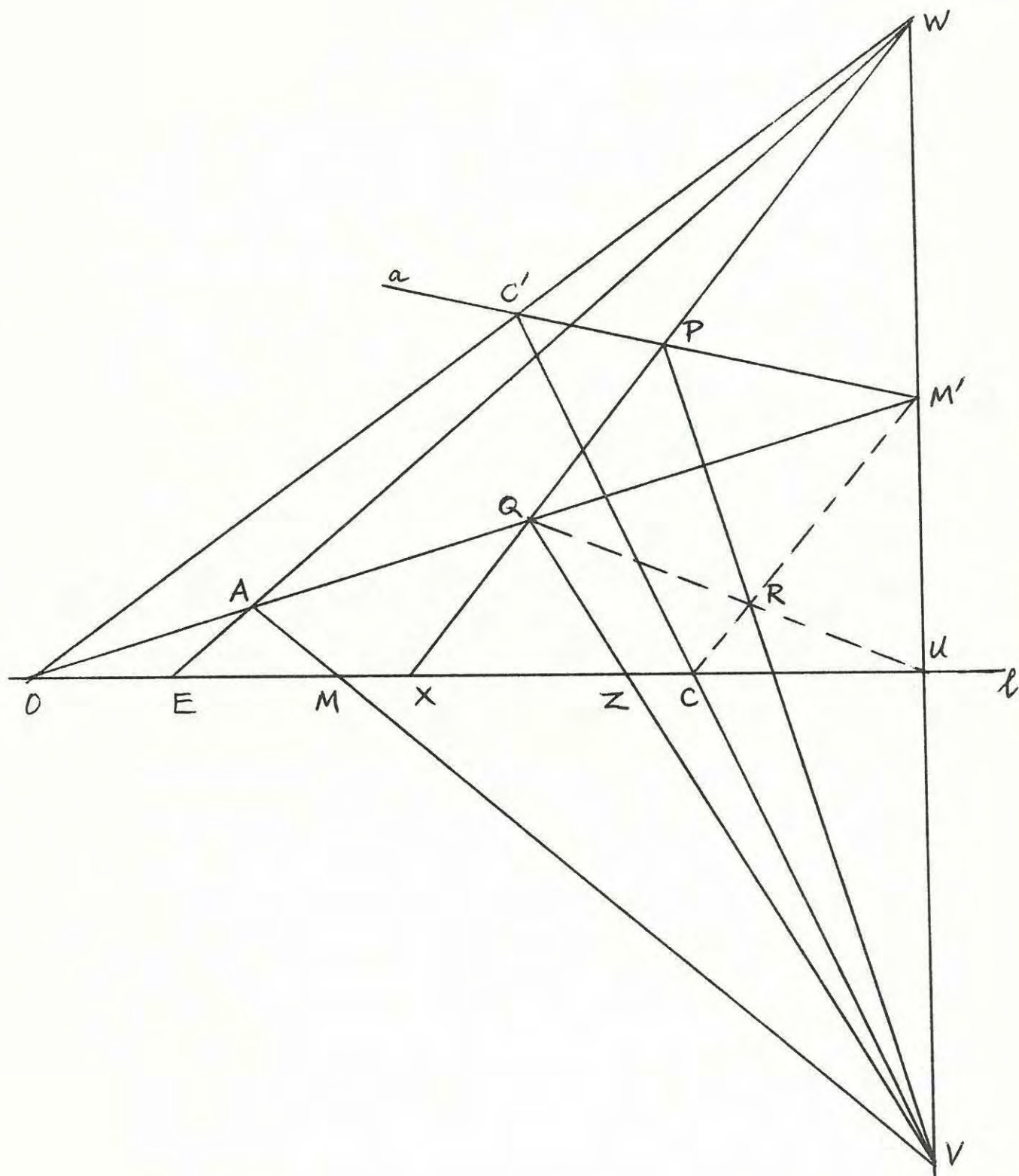
$$y = \mu x + \gamma.$$

Proof: Let $M' = a \cap \ell_\infty$, $A = (O \vee M') \wedge (E \vee W)$, $M = \ell \wedge (A \vee V)$. Then M on ℓ corresponds to μ in F . (Note that $M' \neq W$ since $\mu \neq \infty$). Let $C' = a \cap (O \vee W)$, this has coordinates $(0, \gamma)$, and so $C = \ell \wedge (C' \vee V)$ on ℓ corresponds to γ in F . So $a = C' \vee M'$ is completely determined by γ and μ . Let P be any point on a with coordinates (x, y) , and let X, Y be the points on ℓ corresponding to x, y in F ; that is, $X = (P \vee W) \wedge \ell$, $Y = (P \vee V) \wedge \ell$. Construct $Q = (O \vee M') \wedge (P \vee W)$, $Z = (Q \vee V) \wedge \ell$, and let z be the element of F corresponding to Z . By the converse of Desargues's axiom,

$$\begin{array}{c} \triangle Q M' P \\ C' W O \end{array} \stackrel{V}{\sim} \triangle U C V \implies Q \vee U, M' \vee C, P \vee V \text{ concurrent.}$$

Let R be the common point of $Q \vee U, M' \vee C, P \vee V$. Using the quadrangle $V W Q A$ we obtain $O M E \underline{Q} U X Z$, which means $z = \mu x$. Using the quadrangle $V M' R Q$ we obtain $U Z O \underline{Q} U C Y$, which means $y = z + \gamma$. Hence $y = \mu x + \gamma$ as required. //

For a line a as in the proposition, the pair (μ, γ) is called the affine coordinates of the line a with respect to $\triangle O V W$ and E . So we have affine coordinates for every point of the plane except the ones on ℓ_∞ , and we have affine coordinates for every line of the plane except the ones through W .



§2. Projective planes over skew fields

Let F be a skew field.

The *projective plane* $P_2(F)$ is defined as follows. (Recall that, to give a projective plane its points, lines and incidences must be given).

The *points* are represented by α column vectors $\underline{x} = \begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix}$ where x_0, x_1, x_2 are in F ; and, for any non-zero α in F , the vectors $\underline{x}$ and $\alpha \underline{x}$ represent the same point.

The *lines* are represented by β row vectors $\underline{u}^T = (u_0, u_1, u_2)$ where u_0, u_1, u_2 are in F ; and, for any non-zero β in F , the vectors $\underline{u}^T$ and $\beta \underline{u}^T$ represent the same line.

The point represented by $\underline{x}$ is incident with the line represented by $\underline{u}^T$ when the equation

$$\underline{u}^T \underline{x} = u_0 x_0 + u_1 x_1 + u_2 x_2 = 0$$

is satisfied. Notice that, if $\alpha, \beta \neq 0$, then $\underline{u}^T \underline{x} = 0$ if and only if $(\beta \underline{u}^T)(\alpha \underline{x}) = 0$; so incidence does not depend on the choice of vectors representing the point and line.

The reader should verify the axioms to see that indeed $P_2(F)$ is a projective plane. Also he should check that all the previous examples of projective planes are special cases of this. We saw that the Euclidean plane plus a line and the concurrent lines in three dimensions examples give essentially the same plane, and from the algebraic description of this second example it should be clear that this plane is $P_2(F)$ where F is the field of real numbers. The Fano plane is $P_2(F)$ where F is the field of integers modulo 2, and the thirteen point geometry is $P_2(F)$ where F is the integers modulo 3.

Theorem: Every Desarguesian plane is of the form $P_2(F)$ for some skew field F .

Proof: Let O, V, W, E be any four points in the plane no three of which are collinear. We have seen how to find a field F such that the points and lines in the plane may be given affine coordinates over F with respect to $\triangle OVW$ as triangle of reference and E as unit point. Now we give the procedure for obtaining *homogeneous plane coordinates* from affine coordinates.

The homogeneous coordinates of the point with affine coordinates (x, y) are given by any non-zero multiple of the vector $\begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$. The homogeneous coordinates of a point on $\ell_\infty = V \vee W$ (other than W)

representing gradient μ are given by any non-zero multiple of the vector $\begin{pmatrix} 1 \\ \mu \\ 0 \end{pmatrix}$; for W we take non-zero multiples of the vector $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$.

The homogeneous coordinates of the line with affine coordinates (μ, γ) are given by any non-zero multiple of the vector $(\mu, -1, \gamma)$. The homogeneous coordinates of a line through W (other than ℓ_∞) also passing through $(x, 0)$ are given by any non-zero multiple of the vector $(1, 0, -x)$; for ℓ_∞ we take non-zero multiples of the vector $(0, 0, 1)$.

The student should see that there is a 1-1 correspondence between the set of points of the plane and the set of points of $P_2(F)$, and between the set of lines of the plane and the set of lines $P_2(F)$; these correspondences are obtained by taking homogeneous coordinates. Moreover, a point with homogeneous coordinates $\begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix}$ is on a line with homogeneous coordinates (u_0, u_1, u_2) exactly when $u_0x_0 + u_1x_1 + u_2x_2 = 0$. For example, $\begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$ is on $(\mu, -1, \gamma)$ when $y = \mu x + \gamma$; and $\begin{pmatrix} 1 \\ \mu \\ 0 \end{pmatrix}$ is on $(\mu, -1, \gamma)$. //